

Compal Confidential

ZIWB2/ZIWB3/ZIWE1 DIS M/B Schematics Document

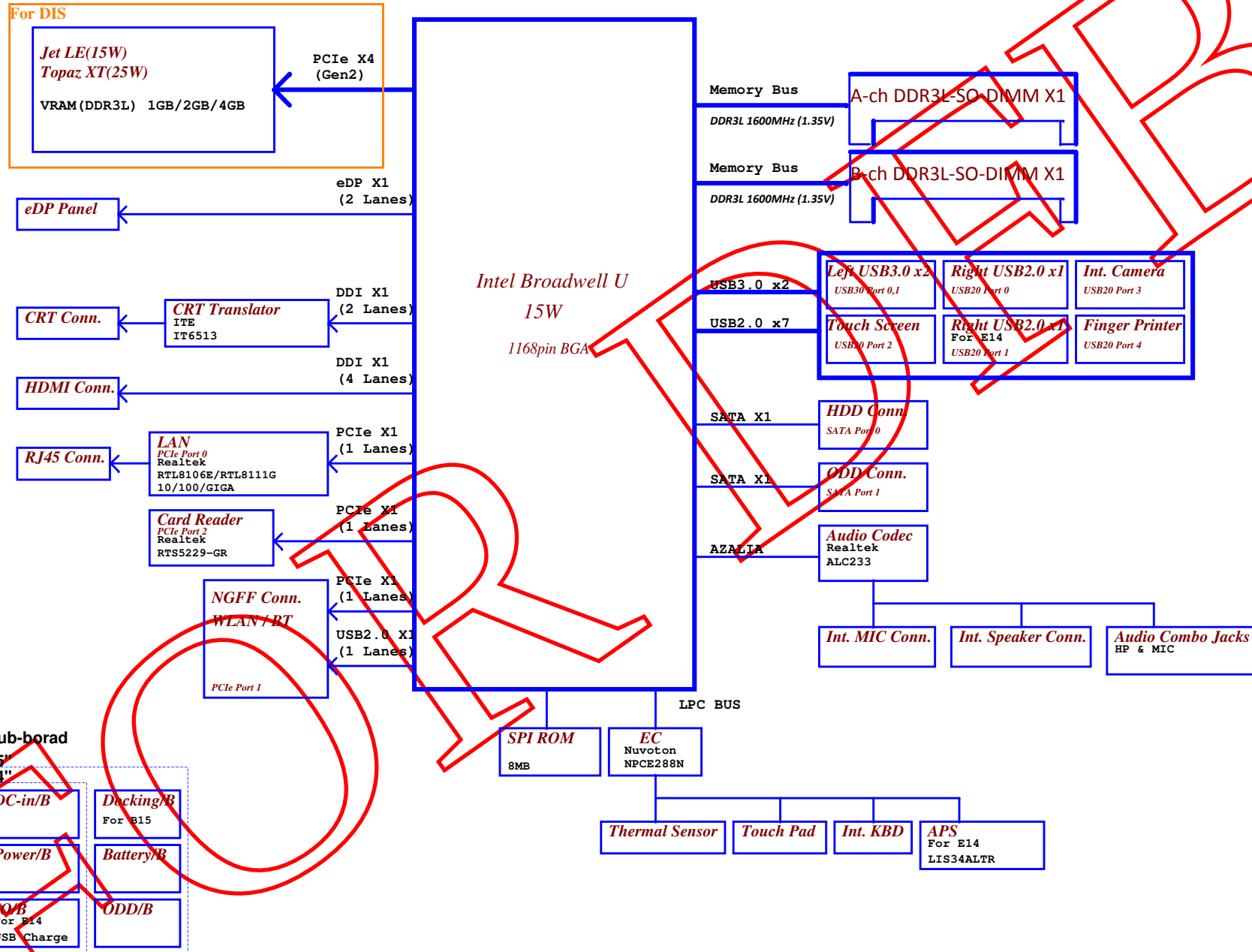
Intel Boardwell U Processor with DDR3L
AMD Topaz XT / Jet LE

2014-02-10

LA-B091P

REV : 1.0

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Issued Date	2011/06/24	Deciphered Date	2012/07/12	Title		
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				C	LA-B091P	1.0
				Date:	Wednesday, February 12, 2014	Sheet 1 of 55



Voltage Rails

power plane	+B	+5VALW	+1.5V	+5VS +3VS +1.5VS +V1.05S_VCCP +VCC_CORE +VGA_CORE +VCC_GFXCORE_AXG +1.8VS +0.75VS +1.05VS
State				
S0	○	○	○	○
S3	○	○	○	×
S5 S4/AC	○	○	×	×
S5 S4/ Battery only	○	×	×	×
S5 S4/AC & Battery don't exist	×	×	×	×

USB Port Table

	USB 2.0 Port	3 External USB Port
EHCI1	UHCI0	0 USB Port (Left Side) USB3.0
		1 USB Port (Left Side) USB3.0
	UHCI1	2 Touch Screen
		3 Camera
	UHCI2	4
		5
EHCI2	UHCI3	6
		7
	UHCI4	8
		9 USB Port (Right Side USB-BD)
	UHCI5	10 Mini Card(WLAN)
		11 Card Reader
	UHCI6	12
		13

BOM Structure Table

Item	BOM Structure
ZIWB2 (14")	B14@
ZIWB3 (15")	B15@
ZIWE1 (14")	B14@
CPU_SA00006SM20	33_42000@
CPU_SA00007AM00	QFS1@
CPU_SA00006SU30	i3_4100U@
CPU_SA000072010	i3_4000U@
CPU_SA00006SY20	i3_4010U@
LAN 10/100 Transformer	100@
LAN GIGB Transformer	GIGA@
LAN Switch mode	SWITCH@
LAN RTL8106E-CG	8106ELDO@
LAN RTL8111GS-CG	8111GLDO@
LAN RTL8106EUS-CG	8106ESW@
LAN RTL8111EUS-CG	8111GSW@
Audio 203	233@
Audio 233VB	233VB@
For B15	Docking@
For B14, E14	NoDocking@
For Deep Sleep	DS3@
For No Deep Sleep	NoDS3@
WLAN Support ISCT	ISCT@
WLAN No Support ISCT	NoISCT@
For Intel ZERO ODD	ZODD@
For No Intel ZERO ODD	NoZODD@
For Green CLK	GCLK@
For No Green CLK	NoGCLK@
For No Green CLK	NoGCLKDIS@
Green CLK IC For DIS	GCLKDIS@
Green CLK IC For UMA	GCLKUMA@
GPU support Dual Rank	DR@
GPU Jet LE	JET@
GPU Topaz XT	TOPAZ@
For DIS	PX@
For UMA	UMA@
Camera	COMS@
APS (G-sensor)	GS@
Touch Screen	TS@
HDMI	HDMI@
USB 2.0	USB2@
USB 3.0	USB3@
Full HD Panel (2 Lane)	FHD@
ENE EC 9012	9012@
HDMI Royalty	45@
Connector	ME@
VRAM indentify	X76@
Un-pop component for EMI	@EMI@
Un-pop component for ESD	@ESD@
DA600140000	PCB_14_DIS@
DA600141000	PCB_14_UMA@
DA600140100	PCB_15_DIS@
DA600141100	PCB_15_UMA@

EC SM Bus1 address

Device	Address	Device	Address
Smart Battery	0001 011x	Thermal Sensor	0100 1100

EC SM Bus2 address

PCH SM Bus address

Device	Address	Device	Address
DDR_IDIMM1	1010 000x A0h	Internal thermal sensor	0100 0001 41h
DDR_IDIMM2	1010 010x A4h		

AMD-GPU SM Bus address

SMBUS Control Table

	SOURCE	VGA	BATT	KB9012	SODIMM	WLAN	Thermal Sensor	PCH
SMB_EC_CK1	KB9012	X	X	X	X	X	X	X
SMB_EC_DA1	+3VALW	+3VALW	X	X	X	X	X	X
SMB_EC_CK2	KB9012	X	X	X	X	X	X	X
SMB_EC_DA2	+3VS	+3VGS	X	X	X	X	X	X
PCH_SMBCLK	PCH	X	X	X	X	X	X	X
PCH_SMBDATA	+3VALW	+3VALW	X	X	X	X	X	X
PCH_SMLCLK	PCH	X	X	X	X	X	X	X
PCH_SMLDATA	+3VALW	+3VALW	X	X	X	X	X	X
SMLCLK	PCH	X	X	X	X	X	X	X
SMLDATA	+3VALW	+3VGS	X	X	X	X	X	X

STATE	SIGNAL	SLP_S1#	SLP_S2#	SLP_S4#	SLP_S5#	+VALW	+V	+VS	Clock
Full ON	HIGH	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	ON
S1 (Power On Suspend)	LOW	HIGH	HIGH	HIGH	HIGH	ON	ON	ON	LOW
S3 (Suspend to RAM)	LOW	LOW	HIGH	HIGH	HIGH	ON	ON	OFF	OFF
S4 (Suspend to Disk)	LOW	LOW	LOW	HIGH	HIGH	ON	OFF	OFF	OFF
S5 (Soft OFF)	LOW	LOW	LOW	LOW	LOW	ON	OFF	OFF	OFF

Topaz XT_VRAM_STRAP

		X76@				X76@			
		Vendor UV5, UV6, UV7, UV8	ID	PS_3[3]	PS_3[2]	PS_3[1]	R_pu RV21	R_pd RV24	
2GBytes	ZZZ01 TH2G@	Hynix 4096Mbits SA00006E800 256Mx16 H5TC4G63AFR-11C	0	0	0	0	NC	4.75K	
1GBytes	ZZZ02 TS1G@	Samsung 2048Mbits SA000068U40 128Mx16 K4W2G1646Q-BC1A	1	0	0	1	8.45K	2K	
1GBytes	ZZZ03 TM1G@	Micron 2048Mbits SA000067500 128Mx16 MT41J128M16JT-093G:K	2	0	1	0	4.53K	2K	
1GBytes	ZZZ04 TH1G@	Hynix 2048Mbits SA00006H400 128Mx16 H5TC2G63FFR-11C	3	0	1	1	6.98K	4.99K	
2GBytes	ZZZ05 TM2G@	Micron 4096Mbits SA000077K00 256Mx16 MT41J256M16HA-093G:E	4	1	0	0	4.53K	4.99K	
2GBytes	ZZZ06 TS2G@	Samsung 4096Mbits SA000076P00 256Mx16 K4W4G1646D-BC1A	5	1	0	1	3.24K	5.62K	
1GBytes	ZZZ07 TM1G2@	Micron 2048Mbits SA00005XB00 128Mx16 MT41K128M16JT-107G:K	6	1	1	0	3.4K	10K	
2GBytes	ZZZ08 TM2G2@	Micron 4096Mbits SA000065D00 256Mx16 MT41K256M16HA-107G:E	7	1	1	1	4.75K	NC	

ZZZ01

ZZZ02

ZZZ03

ZZZ04

TH2G@
2G HYNIX
X7653638L01

TS1G@
1G SAMSUNG
X7653638L03

TM1G@
1G MICRON
X7653638L02

TH1G@
1G HYNIX
X7653638L01

Power-Up/Down Sequence

"Mars" has the following requirements with regards to power-supply sequencing to avoid damaging the ASIC:

- All the ASIC supplies must reach their respective nominal voltages within 20 μ s of the start of the ramp-up sequence, though a shorter ramp-up duration is preferred. The maximum slew rate on all rails is 50 mV/ μ s.
- The external pull ups on the DDC/AUX signals (if applicable) should ramp up before or after both VDDC and VDD_CT have ramped up.
- VDDC and VDD_CT should not ramp up simultaneously. For example, VDDC should reach 90% before VDD_CT starts to ramp up (or vice versa).
- For power down, reversing the ramp-up sequence is recommended.

VDDR3(3.3VGS)

PCIE_VDDC(0.95VGSV)

VDDR1(1.5VGS)

VDDC/VDDC(1.12V)

VDD_CT(1.8V)

PERST#

REFCLK

Straps Reset

Straps Valid

Global ASIC Reset

T4+16clock

Jet LE_VRAM_STRAP

		X76@				X76@			
		Vendor UV5, UV6, UV7, UV8	ID	PS_3[3]	PS_3[2]	PS_3[1]	R_pu RV21	R_pd RV24	
1GBytes	ZZZ09 JH1G@	Hynix 2048Mbits SA00006H400 128Mx16 H5TC2G63FFR-11C	0	0	0	0	NC	4.75K	
1GBytes	ZZZ10 JM1G@	Micron 2048Mbits SA000067500 128Mx16 MT41J128M16JT-093G:K	1	0	0	1	8.45K	2K	
1GBytes	ZZZ11 JS1G@	Samsung 2048Mbits SA000068U40 128Mx16 K4W2G1646Q-BC1A	2	0	1	0	4.53K	2K	
2GBytes	ZZZ12 JH2G@	Hynix 4096Mbits SA00006E800 256Mx16 H5TC4G63AFR-11C	3	0	1	1	6.98K	4.99K	
2GBytes	ZZZ13 JS2G@	Samsung 4096Mbits SA000076P00 256Mx16 K4W4G1646D-BC1A	4	1	0	0	4.53K	4.99K	
2GBytes	ZZZ14 JM2G@	Micron 4096Mbits SA000077K00 256Mx16 MT41J256M16HA-093G:E	5	1	0	1	3.24K	5.62K	
2GBytes	ZZZ08 JM2G2@	Micron 4096Mbits SA000065D00 256Mx16 MT41K256M16HA-107G:E	6	1	1	0	3.4K	10K	
1GBytes	ZZZ16 JM1G2@	Micron 2048Mbits SA00005XB00 128Mx16 MT41K128M16JT-107G:K	7	1	1	1	4.75K	NC	

ZZZ09

ZZZ10

ZZZ11

ZZZ12

JH1G@
1G HYNIX
X7653638L07

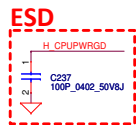
JM1G@
1G MICRON
X7653638L09

JS1G@
1G SAMSUNG
X7653638L09

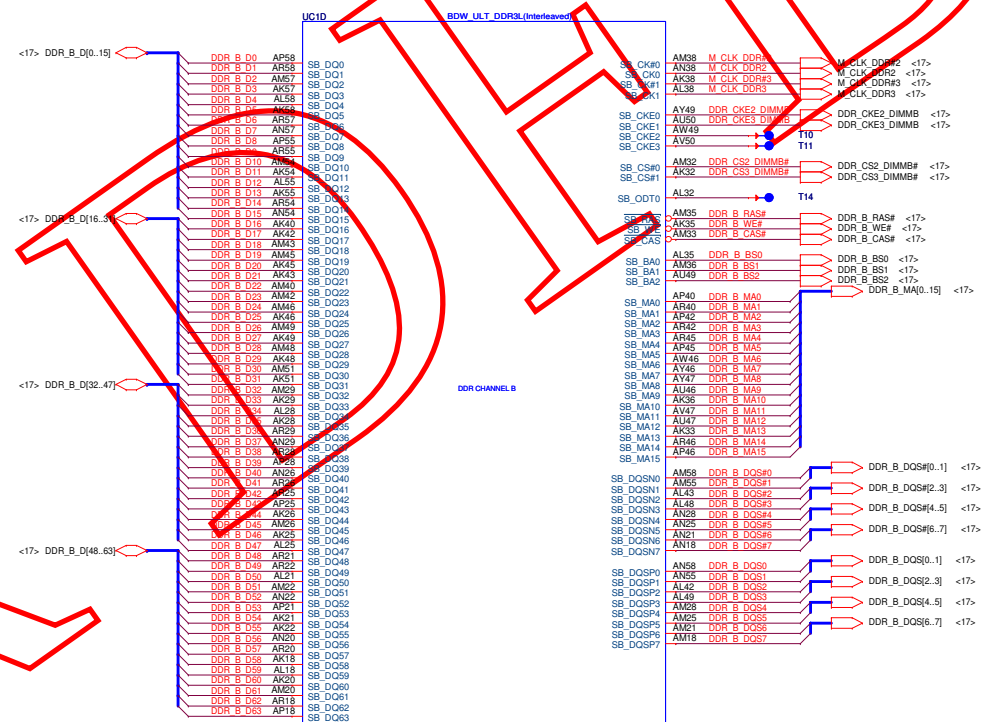
JH2G@
2G HYNIX
X7653638L04

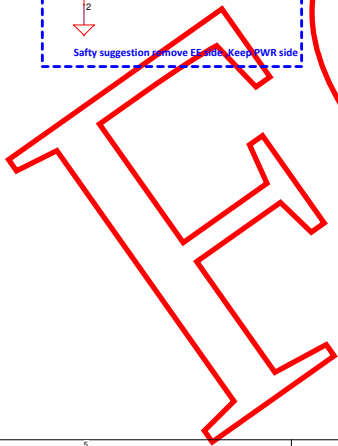
R_pu (Ω)	R_pd (Ω)	Bits [3:1]
NC	4750	000
8450	2000	001
4530	2000	010
6980	4990	011
4530	4990	100
3240	5620	101
3400	10000	110
4750	NC	111

Note: 0402 1% resistors are required.

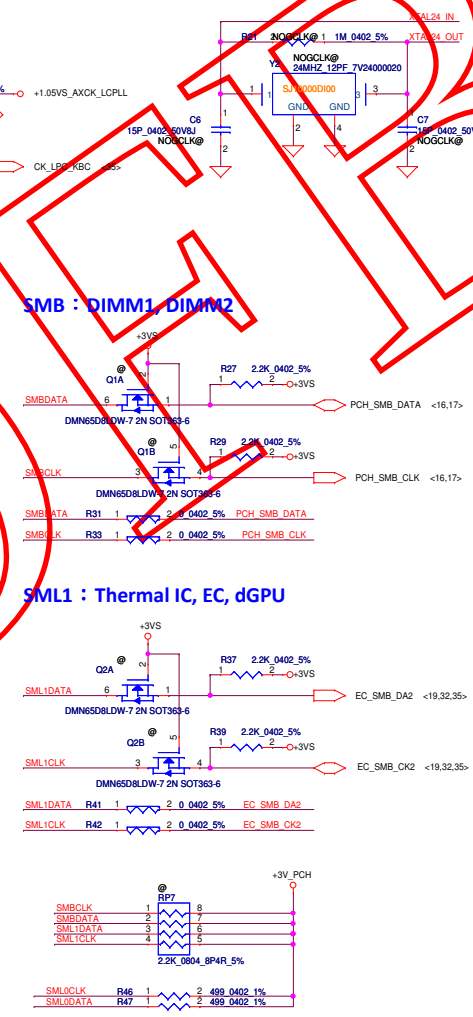


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Sheet 5 of 55 |



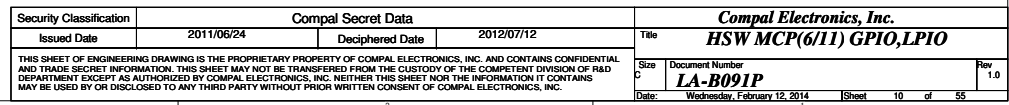


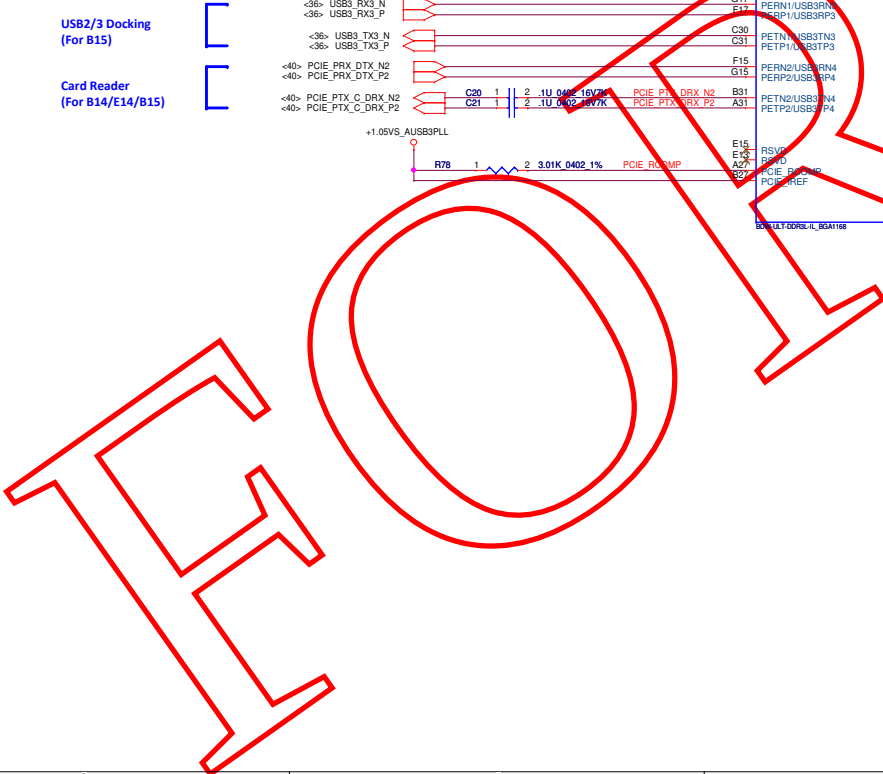
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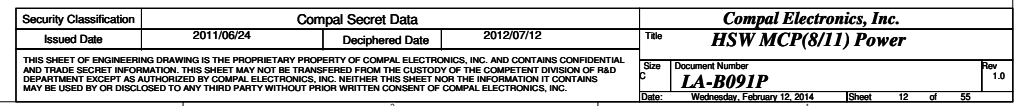
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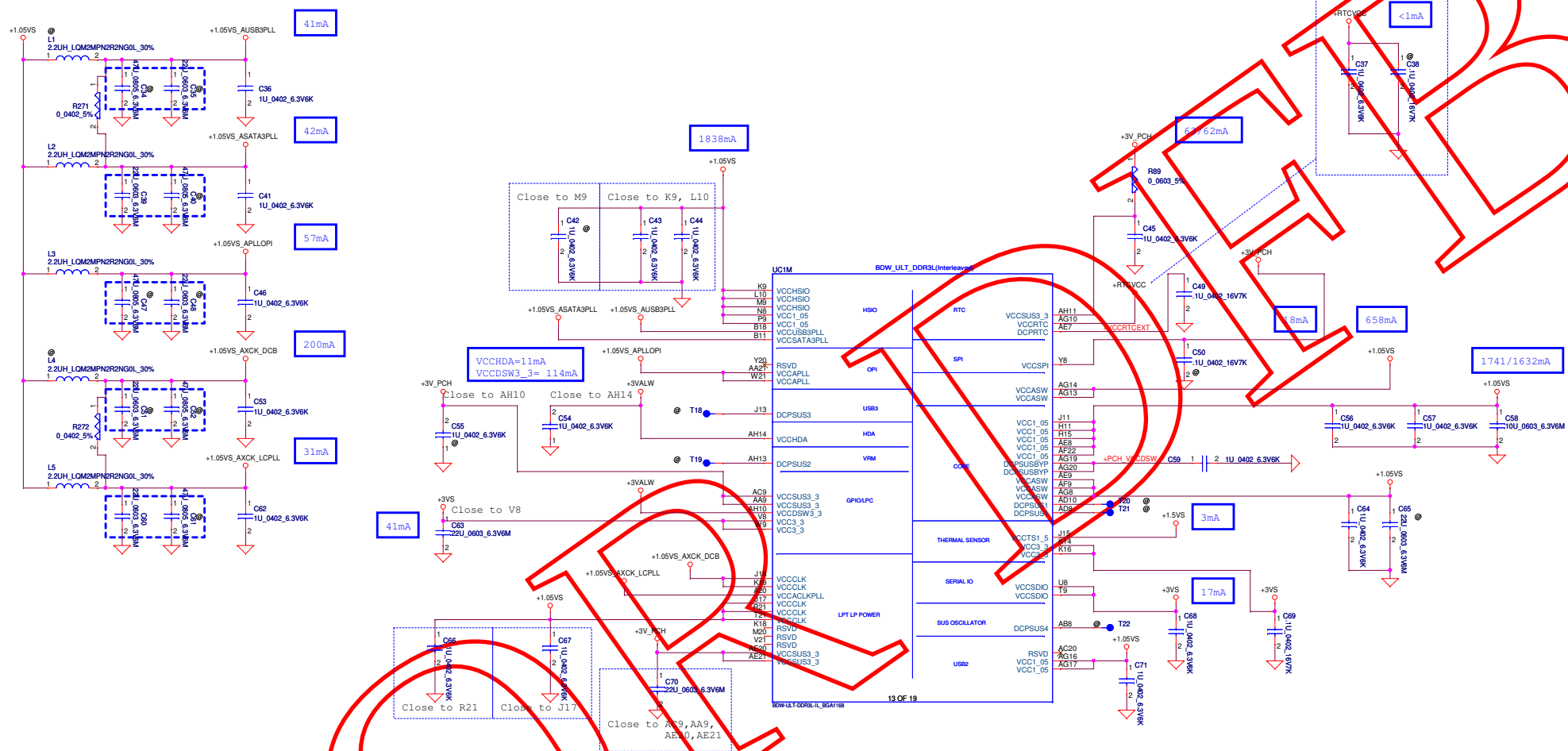
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				C	LA-B091P	1.0
				Date	Wednesday, February 12, 2014	Sheet 9 of 55



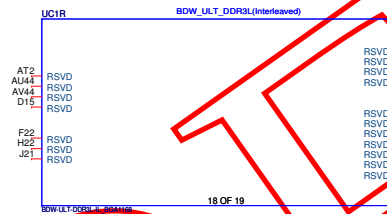


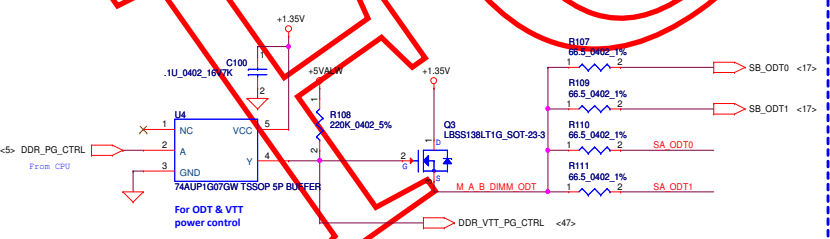
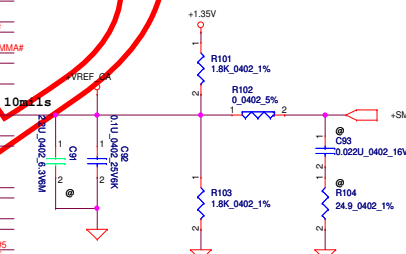
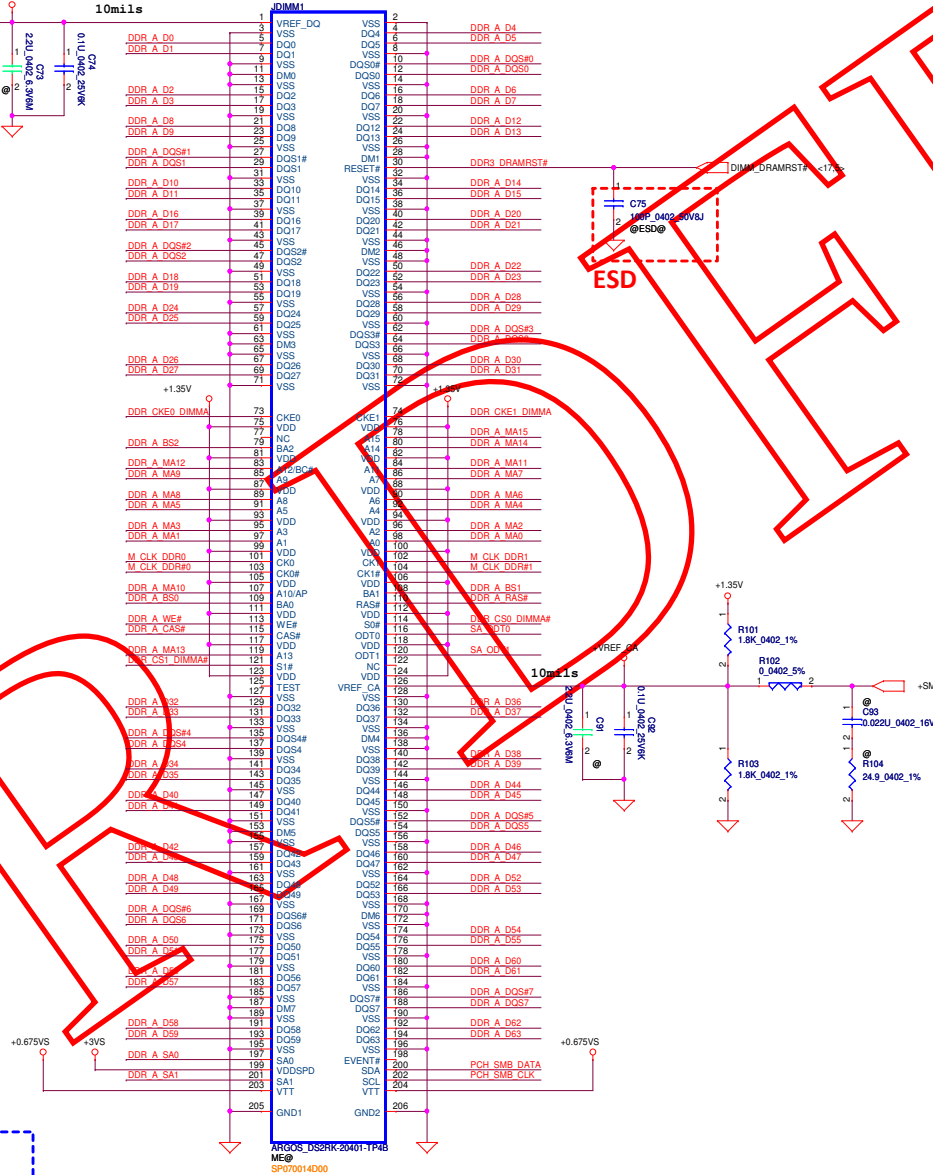
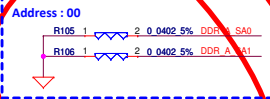
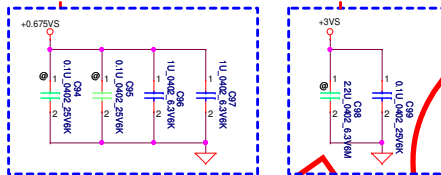
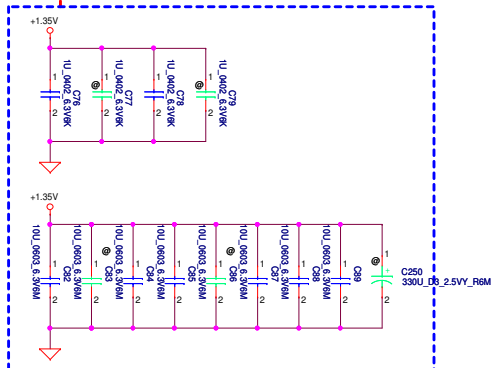
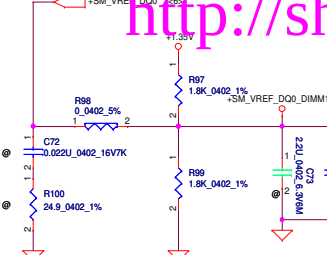
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UC1N BDW ULT_DDR3L(Interleaved)			UC1O BDW ULT_DDR3L(Interleaved)			UC1P BDW ULT_DDR3L(Interleaved)		
A11	VSS	AJ35	AP22	VSS	AV59	D33	VSS	H11
A14	VSS	AJ39	AP23	VSS	AV8	D34	VSS	H12
A18	VSS	AJ41	AP26	VSS	AW16	D35	VSS	J10
A24	VSS	AJ43	AP29	VSS	AW24	D37	VSS	J22
A28	VSS	AJ45	AP3	VSS	AW33	D38	VSS	J39
A32	VSS	AJ47	AP31	VSS	AW35	D39	VSS	J63
A36	VSS	AJ50	AP38	VSS	AW37	D41	VSS	K12
A40	VSS	AJ52	AP39	VSS	AW4	D42	VSS	K1
A44	VSS	AJ54	AP48	VSS	AW40	D43	VSS	L15
A48	VSS	AJ55	AP52	VSS	AW42	D45	VSS	L17
A52	VSS	AJ58	AP54	VSS	AW44	D46	VSS	L18
A56	VSS	AJ60	AP57	VSS	AW47	D47	VSS	L20
A61	VSS	AJ63	AR11	VSS	AW50	D49	VSS	L58
AX58	VSS	AJ65	AR15	VSS	AW51	D5	VSS	L61
AB10	VSS	AK3	AR17	VSS	AW52	D50	VSS	L7
AB20	VSS	AK52	AR23	VSS	AW59	D51	VSS	M22
AB23	VSS	AL10	AR31	VSS	AY11	D53	VSS	N10
AB7	VSS	AL13	AR33	VSS	AY22	D54	VSS	N3
AD21	VSS	AL17	AR39	VSS	AY24	D55	VSS	P59
AD3	VSS	AL21	AR43	VSS	AY26	D57	VSS	P83
AD5	VSS	AL22	AR49	VSS	AY28	D59	VSS	R10
AE10	VSS	AL28	AR5	VSS	AY30	D62	VSS	R22
AE5	VSS	AL29	AT13	VSS	AY33	D8	VSS	R8
AE8	VSS	AL31	AT35	VSS	AY4	E11	VSS	T1
AF11	VSS	AL33	AT37	VSS	AY51	E17	VSS	T58
AF12	VSS	AL36	AT40	VSS	AY53	F20	VSS	U20
AF14	VSS	AL38	AT42	VSS	AY57	F26	VSS	U22
AF15	VSS	AL40	AT43	VSS	AY59	F34	VSS	U61
AF17	VSS	AL43	AT46	VSS	AY6	F38	VSS	U9
AF18	VSS	AL45	AT49	VSS	B20	F40	VSS	Y10
AG1	VSS	AL51	AT61	VSS	B24	F42	VSS	Y3
AG11	VSS	AL52	AT62	VSS	B26	F46	VSS	Y7
AG21	VSS	AL54	AT65	VSS	B28	F50	VSS	W20
AG23	VSS	AL57	AU1	VSS	B32	F53	VSS	W22
AG59	VSS	AL60	AU16	VSS	B36	F61	VSS	Y10
AG61	VSS	AL61	AU18	VSS	B4	G18	VSS	Y83
AG62	VSS	AM1	AU20	VSS	B40	G22	VSS	
AG63	VSS	AM17	AU22	VSS	B44	G3	VSS	
AH17	VSS	AM23	AU24	VSS	B48	G5	VSS	V8
AH19	VSS	AM31	AU26	VSS	B52	G6	VSS	AH46
AH20	VSS	AM34	AU28	VSS	B56	G8	VSS	Y23
AH22	VSS	AN17	AU30	VSS	B60	H13	VSS	E62
AH24	VSS	AN23	AU33	VSS	C11		VSS	AH16
AH26	VSS	AN3	AU35	VSS	C14			
AH30	VSS	AN35	AU37	VSS	C20			
AH32	VSS	AN36	AU39	VSS	C27			
AH34	VSS	AN39	AU57	VSS	C29			
AH38	VSS	AN40	AU59	VSS	C38			
AH40	VSS	AN42	AV15	VSS	C39			
AH42	VSS	AN43	AV20	VSS	C57			
AH44	VSS	AN45	AV23	VSS	D12			
AH49	VSS	AN46	AV28	VSS	D14			
AH51	VSS	AN48	AV33	VSS	D18			
AH53	VSS	AN49	AV34	VSS	D25			
AH55	VSS	AN51	AV36	VSS	D27			
AH57	VSS	AN52	AV39	VSS	D28			
AJ1	VSS	AN53	AV41	VSS	D29			
AJ14	VSS	AN55	AV43	VSS	D26			
AJ23	VSS	AN7	AV45	VSS	D27			
AJ25	VSS	AP10	AV49	VSS	D27			
AJ27	VSS	AP17	AV51	VSS	D30			
AJ29	VSS	AP20	AV55	VSS	D31			





Interleaved Memory

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Author [Blank]				[Blank]	[Blank]



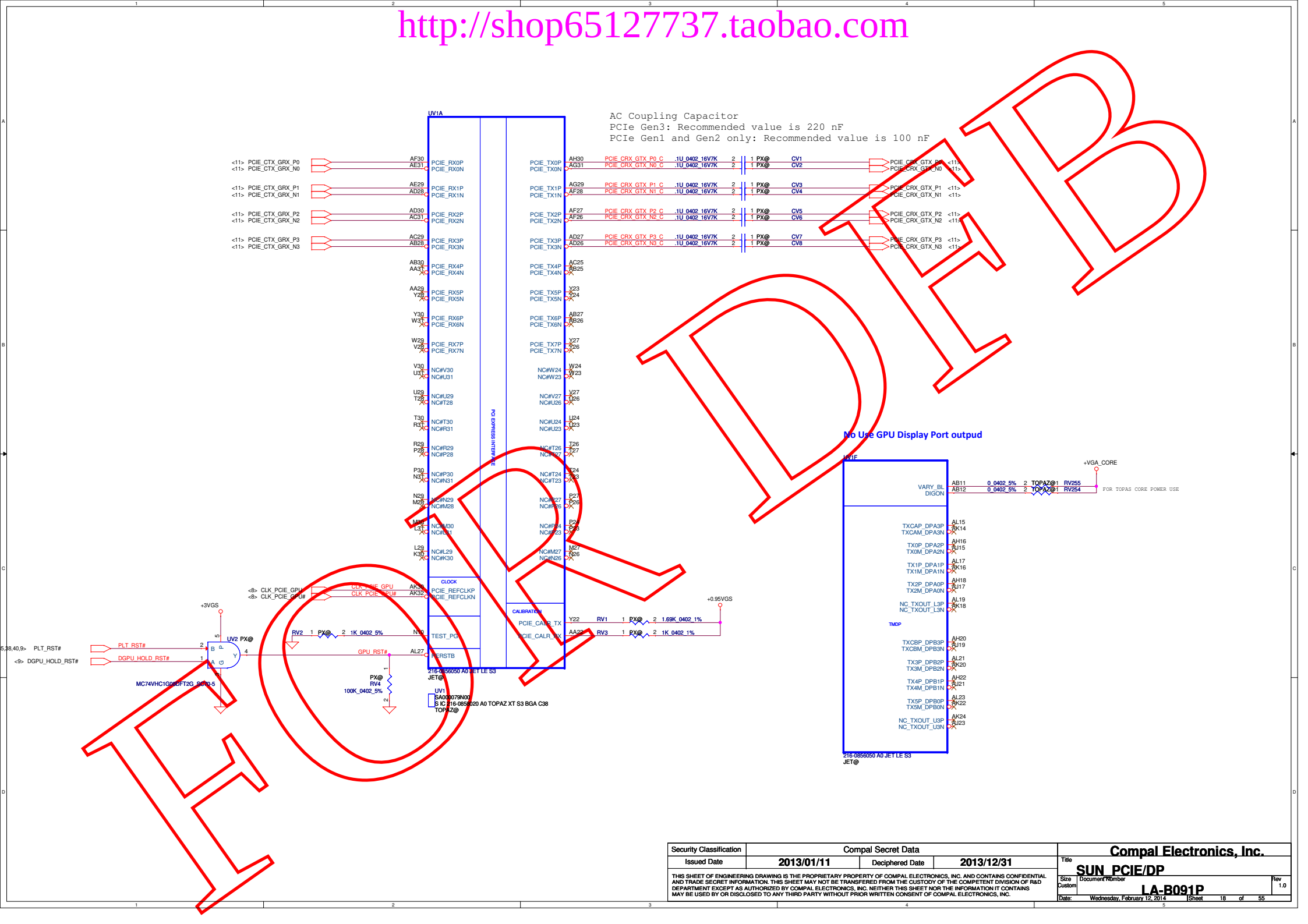
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AC Coupling Capacitor
 PCIe Gen3: Recommended value is 220 nF
 PCIe Gen1 and Gen2 only: Recommended value is 100 nF

UV1A

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 PCI



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AC Coupling Capacitor
PCIe Gen3: Recommended value is 220 nF
PCIe Gen1 and Gen2 only: Recommended value is 100 nF

UV1A

PCIe CRX GTX P0 C .1U 0402 16V7K 2 1 PX@ CV1
PCIe CRX GTX N0 C .1U 0402 16V7K 2 1 PX@ CV2

PCIe CRX GTX P1 C .1U 0402 16V7K 2 1 PX@ CV3
PCIe CRX GTX N1 C .1U 0402 16V7K 2 1 PX@ CV4

PCIe CRX GTX P2 C .1U 0402 16V7K 2 1 PX@ CV5
PCIe CRX GTX N2 C .1U 0402 16V7K 2 1 PX@ CV6

PCIe CRX GTX P3 C .1U 0402 16V7K 2 1 PX@ CV7
PCIe CRX GTX N3 C .1U 0402 16V7K 2 1 PX@ CV8

UV1F

VARY BL DIGON AB11 0.0402 5% 2 TOPAZ@1 RV255
AB12 0.0402 5% 2 TOPAZ@1 RV254 FOR TOPAZ CORE POWER USE

TXCAP_DPA3P AL15
TXCAM_DPA3N AL14
TXOP_DPA2P AH16
TXOM_DPA2N AJ15
TX1P_DPA1P AL17
TX1M_DPA1N AK16
TX2P_DPA0P AH18
TX2M_DPA0N AJ17
NC_TXOUT_L3P AL19
NC_TXOUT_L3N AK18

TXCBP_DPB3P AH20
TXCBM_DPB3N AJ19
TX3P_DPB2P AL21
TX3M_DPB2N AK20
TX4P_DPB1P AH22
TX4M_DPB1N AJ21
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TX5M_DPB0N AK22
NC_TXOUT_L3P AK24
NC_TXOUT_L3N AJ23

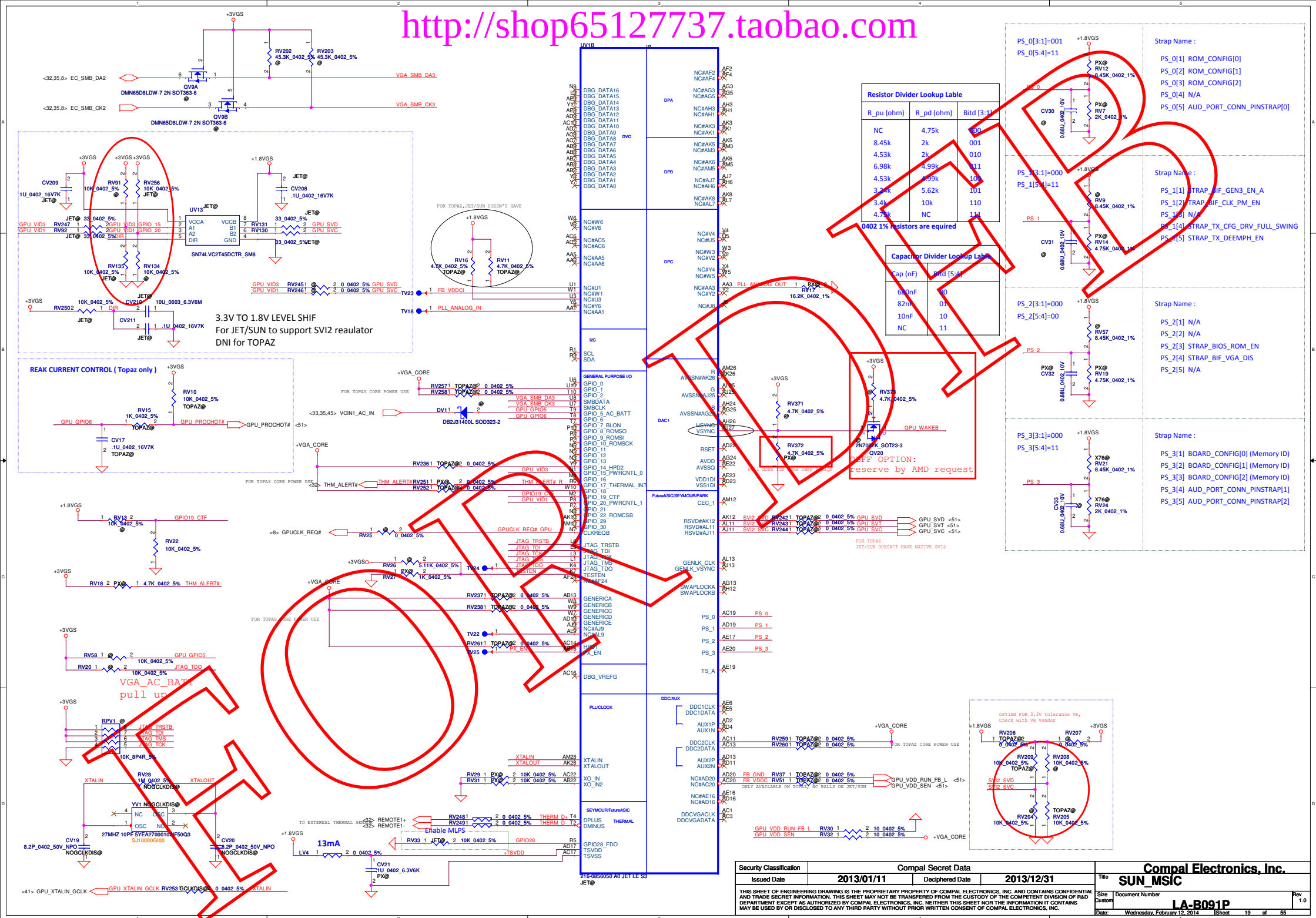
216-0856050 A0 JET LE S3
JET@

Security Classification
Issued Date 2013/01/11
Deciphered Date 2013/12/31

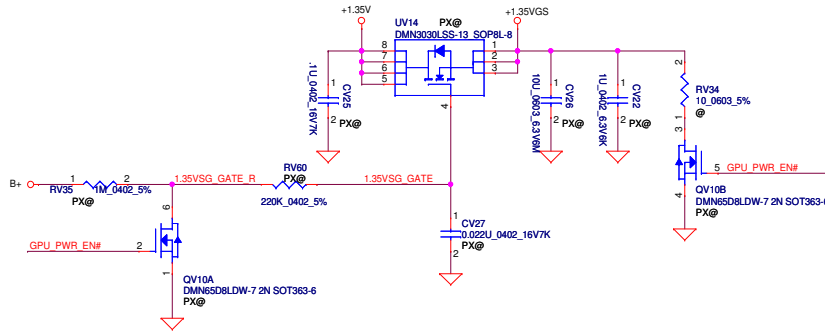
Compal Secret Data

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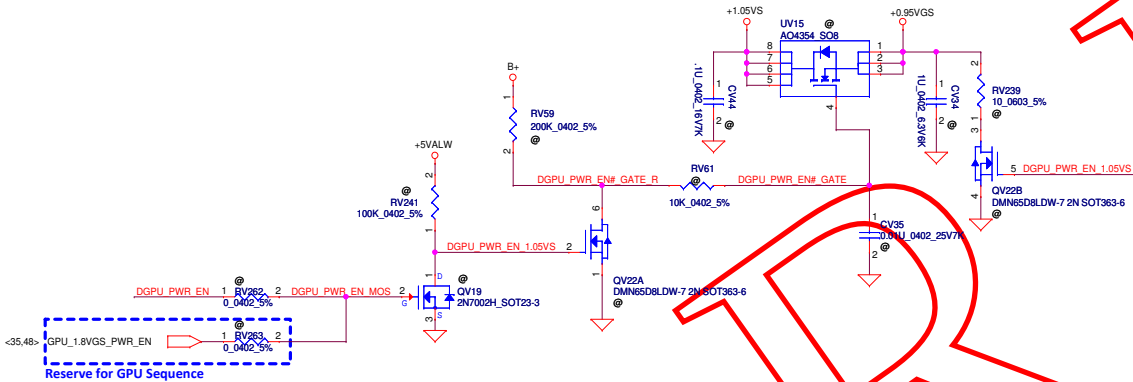
Compal Electronics, Inc.
Title SUN PCIe/DP
Size Custom
Document Number LA-B091P
Date: Wednesday, February 12, 2014
Sheet 18 of 55



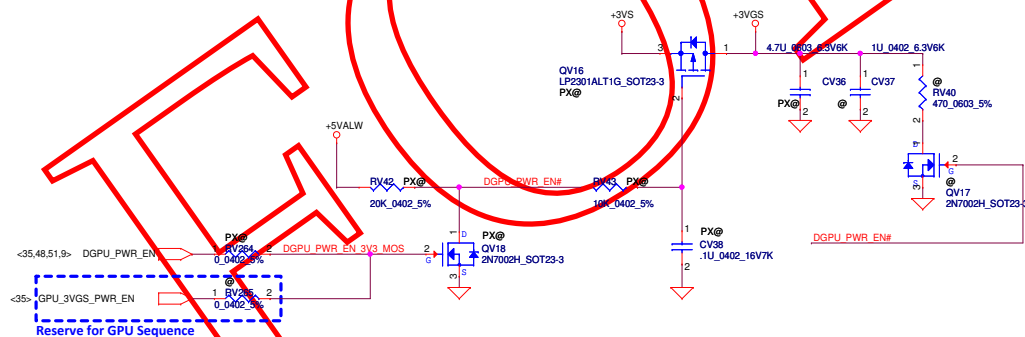
+1.35VS to +1.35VGS (6.234A)



+1.05VS to +0.95VGS



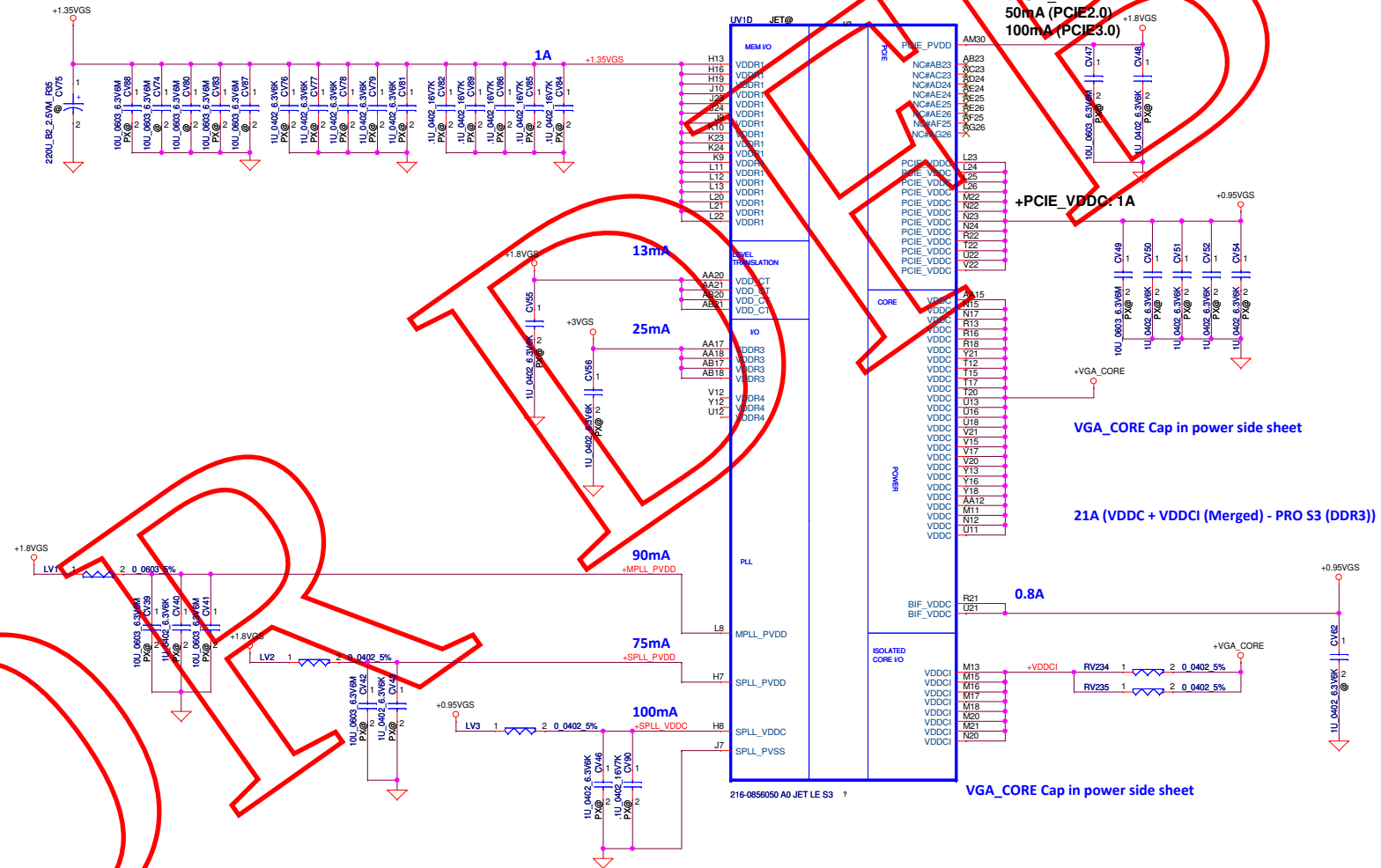
+3VS to +3VS_VGA (25mA)

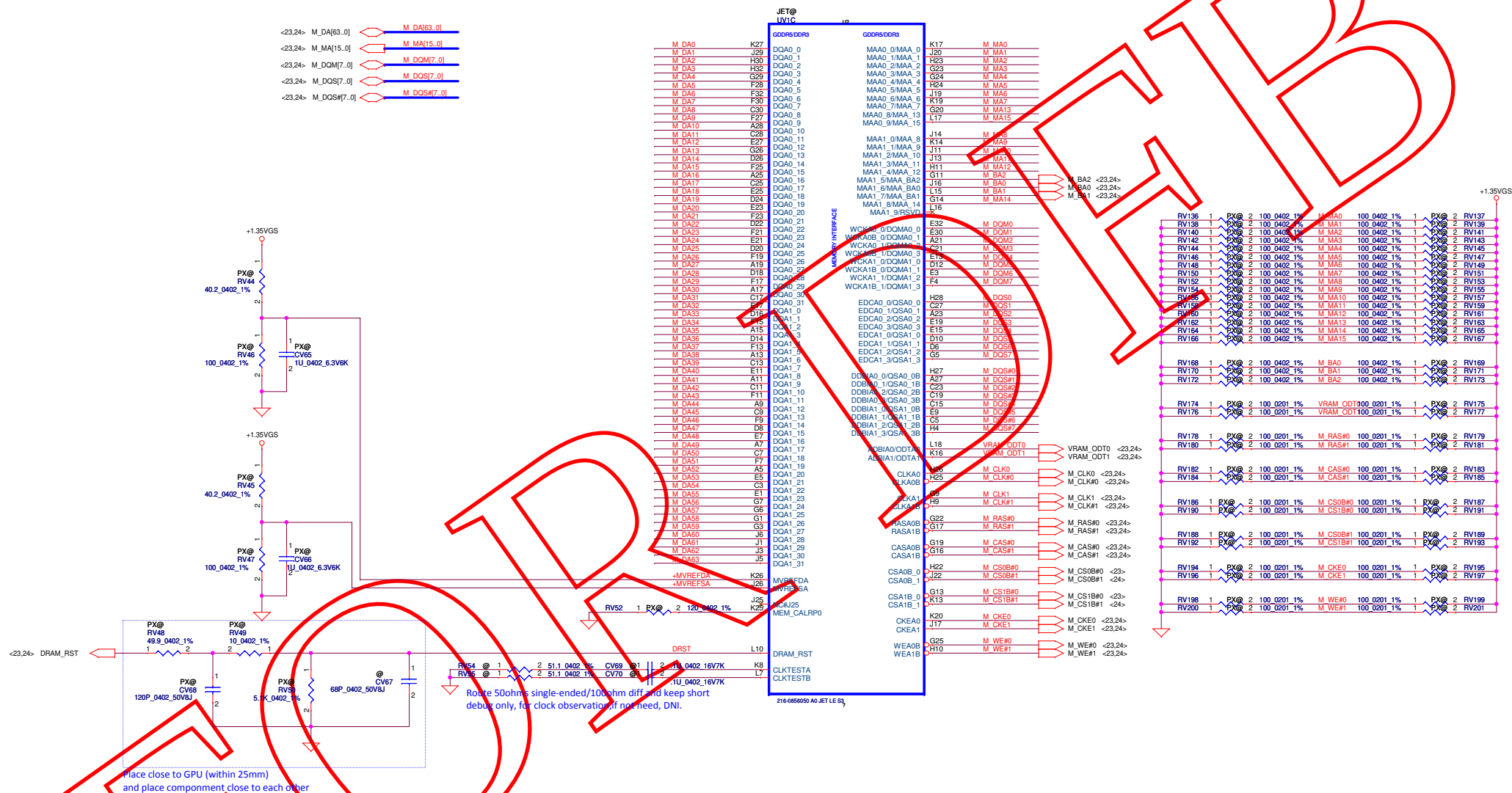


No Use GPU Display Port output

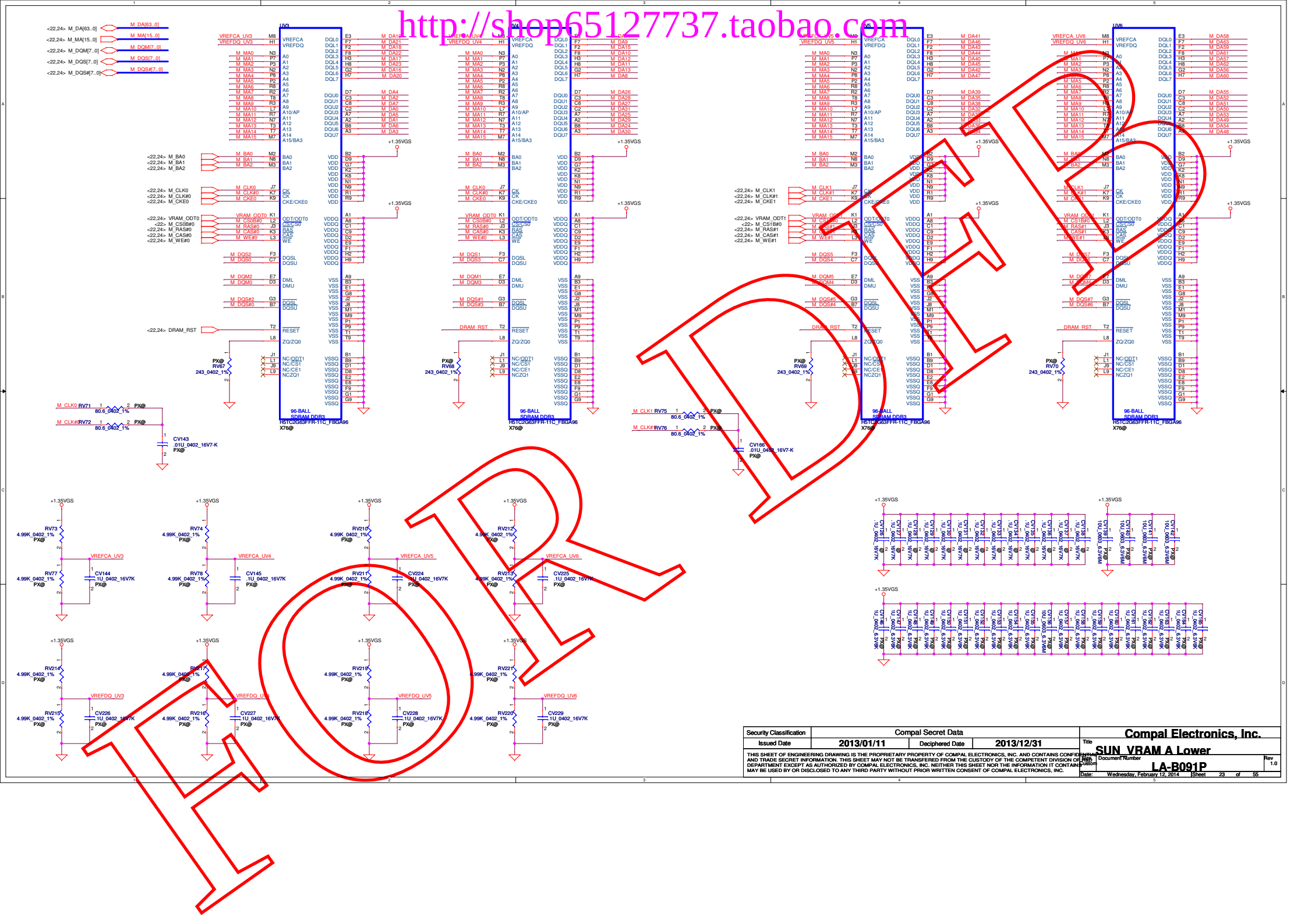


+3VGS		10uF	1uF	0.1uF
VDDR3	25mA	0	1	0



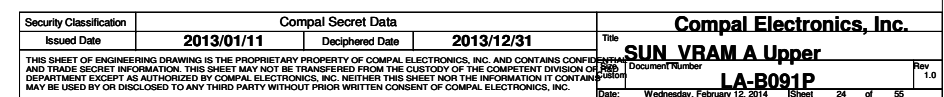


Security Classification	Compal Secret Data			Compal Electronics, Inc.		
Issued Date	2013/01/11	Deciphered Date	2013/12/31	Title	SUN MEM	
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				Custom	LA-B091P	1.0
				Date	Wednesday, February 19, 2014	Page 99 of 66



Security Classification		Compal Secret Data		Compal Electronics, Inc.	
Issued Date	2013/01/11	Deciphered Date	2013/12/31	Title	SUN VRAM A Lower
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				Date	Wednesday, February 12, 2014
				Sheet	23 of 55

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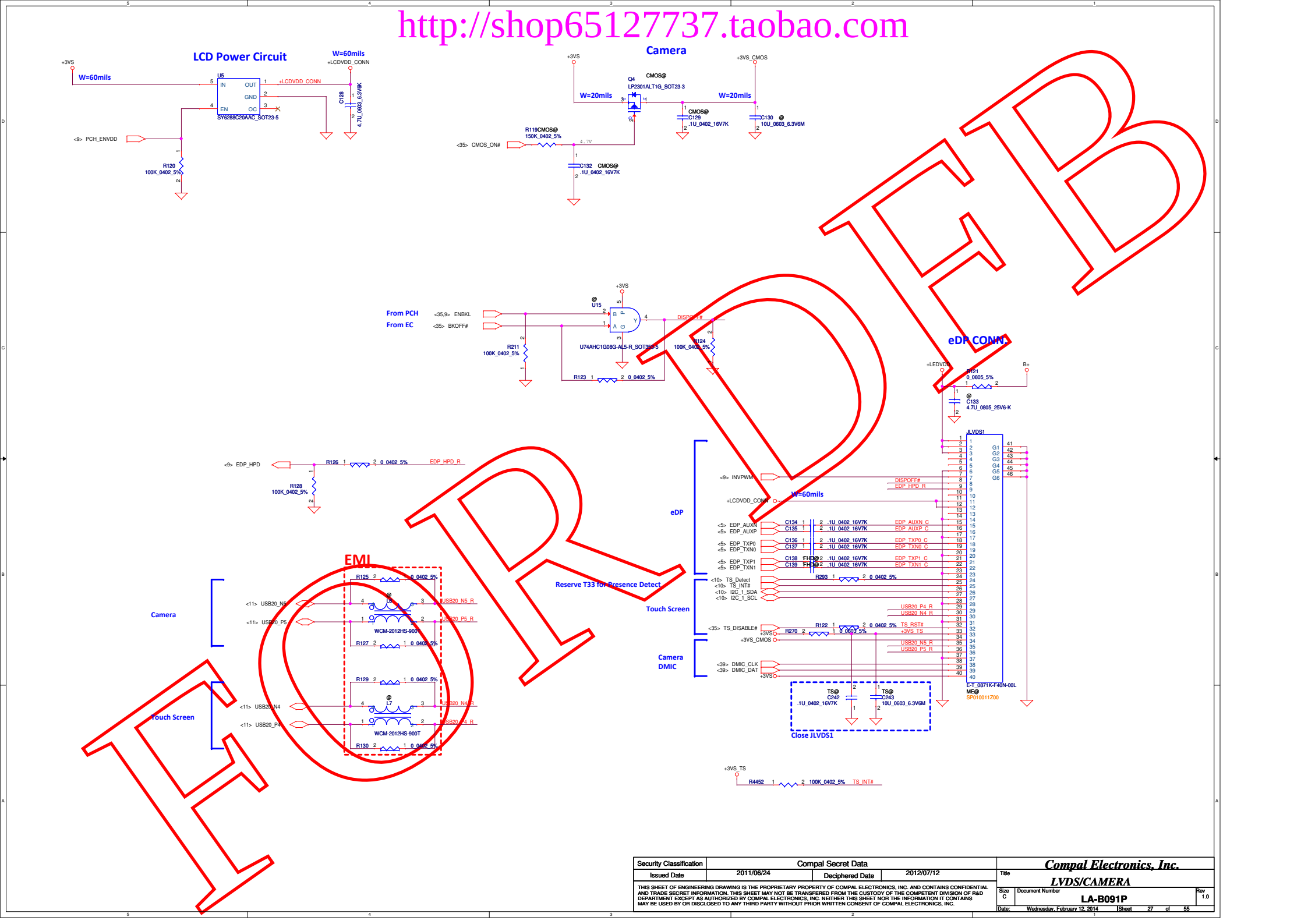


FOR DEB

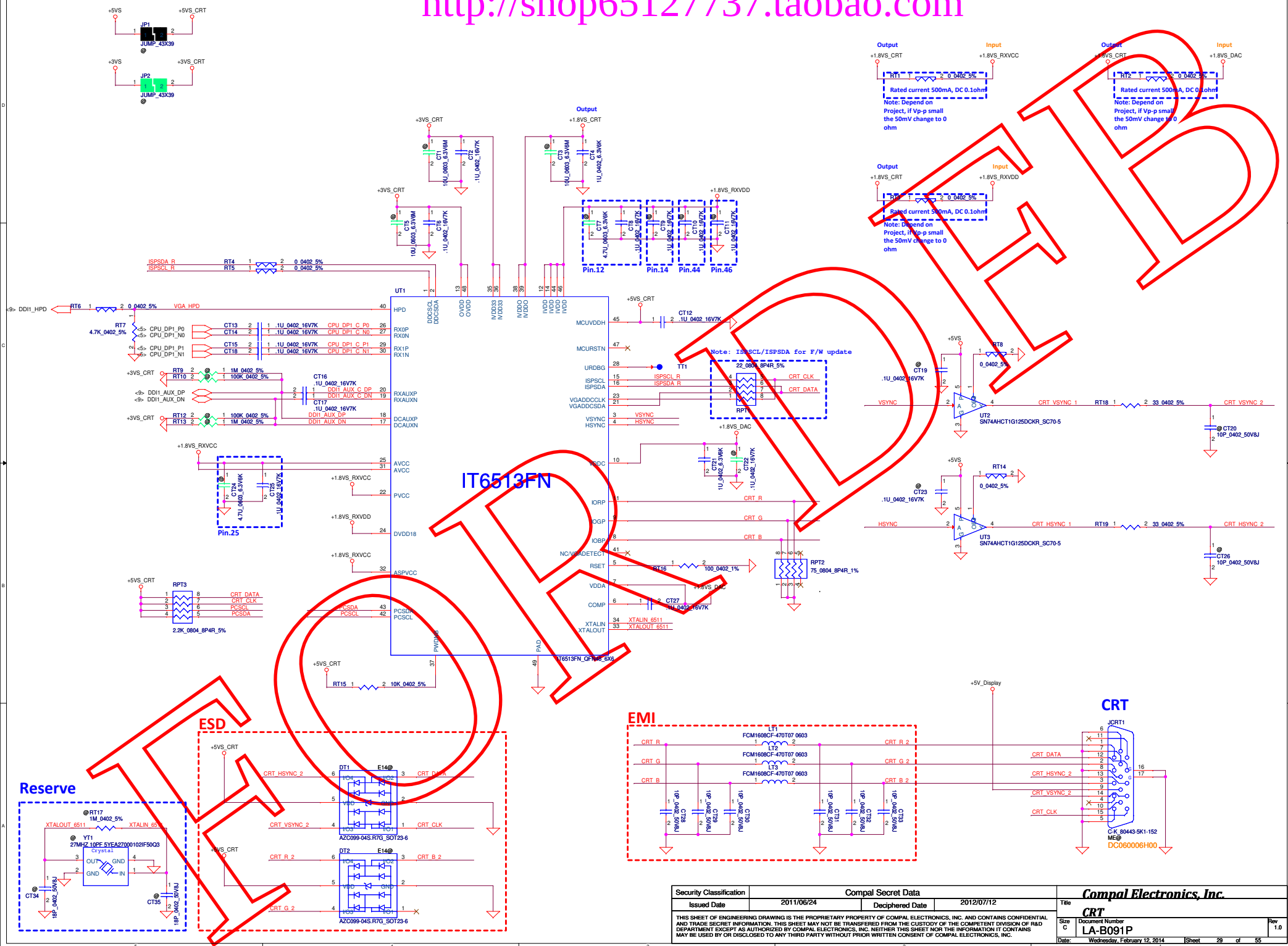
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Issued Date	2013/01/11	Deciphered Date	2013/12/31	SUN VRAM A Lower	
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				Date: Wednesday, February 12, 2014	Sheet 25 of 55

FOR DEB

Security Classification		Compal Secret Data		Title	
Issued Date	2013/01/11	Deciphered Date	2013/12/31	SUN VRAM A Upper	
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				Date: Wednesday, February 12, 2014	Sheet 26 of 55

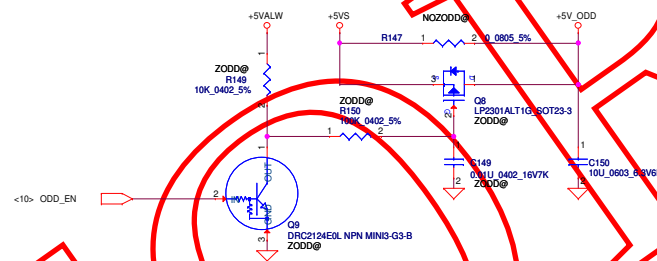
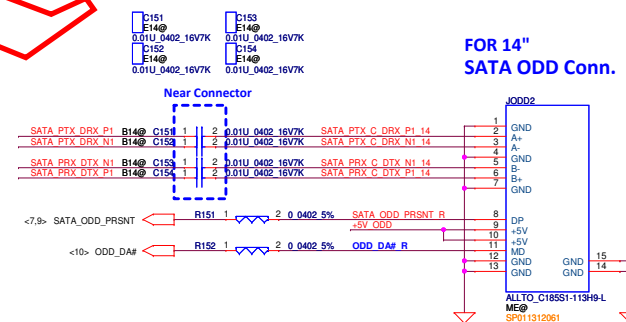
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FOR 15"
SATA ODD FFC Conn.

FOR 14"
SATA ODD Conn.

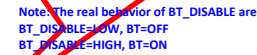


Security Classification		Compal Secret Data		Compal Electronics, Inc. HDD/ODD/BT Connector	
Issued Date	2011/06/24	Deciphered Date	2012/07/12	Title	
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					Rev 1.0
				Date:	Wednesday, February 12, 2014 Sheet 30 of 55

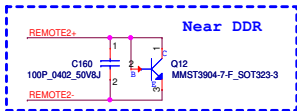
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WLAN

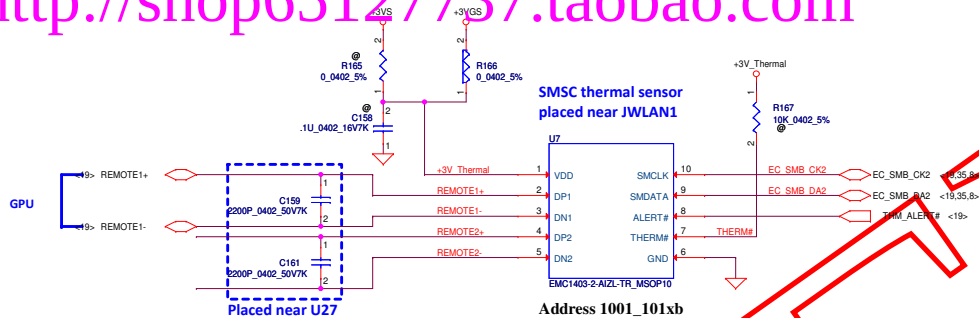
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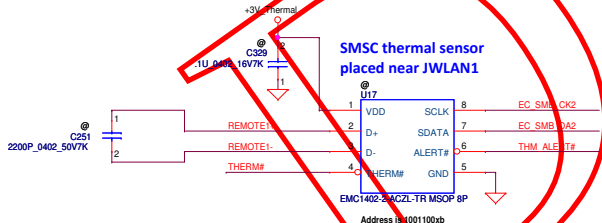
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Issued Date	2011/06/24	Deciphered Date	2012/07/12	Title		
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				Size	Document Number	Rev
					LA-B091P	1.0
Date:				Wednesday, February 12, 2014	Sheet	31 of 55



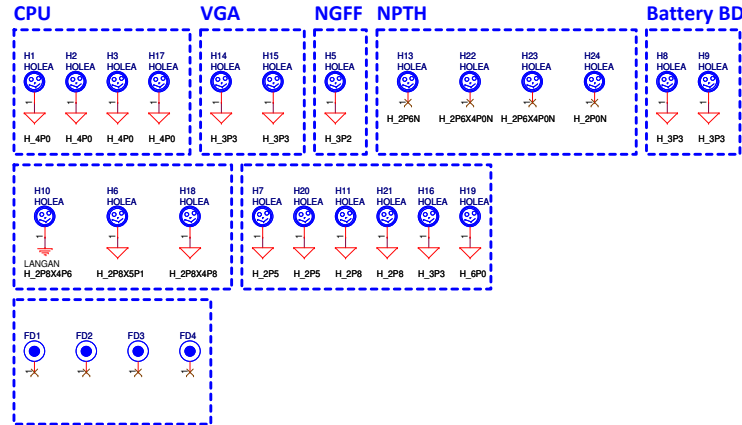
REMOTE1,2+/-:
Trace width/space:10/10 mil
Trace length:<8"



Address 1001_101xb

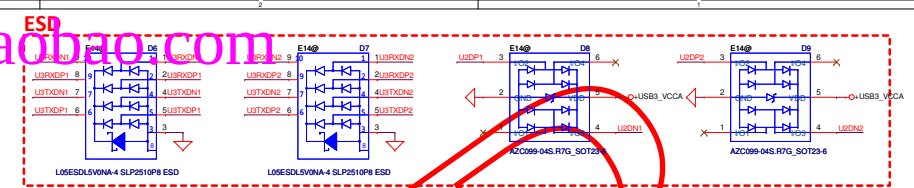
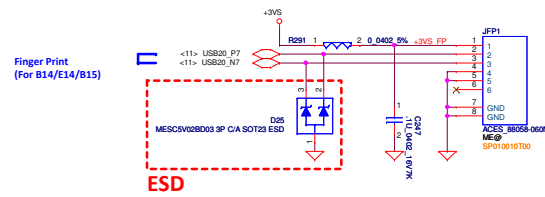


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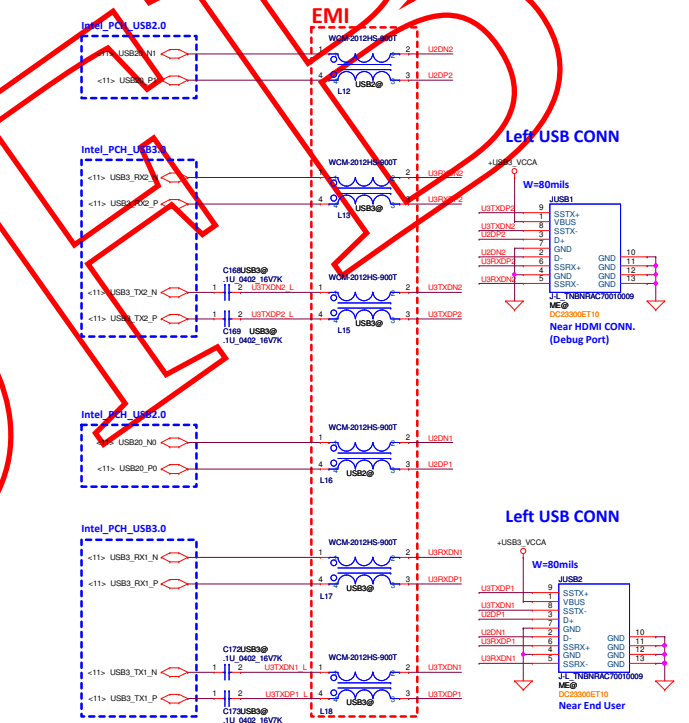


Finger Print

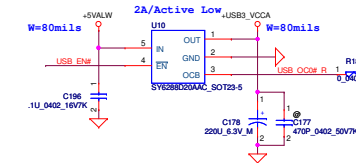
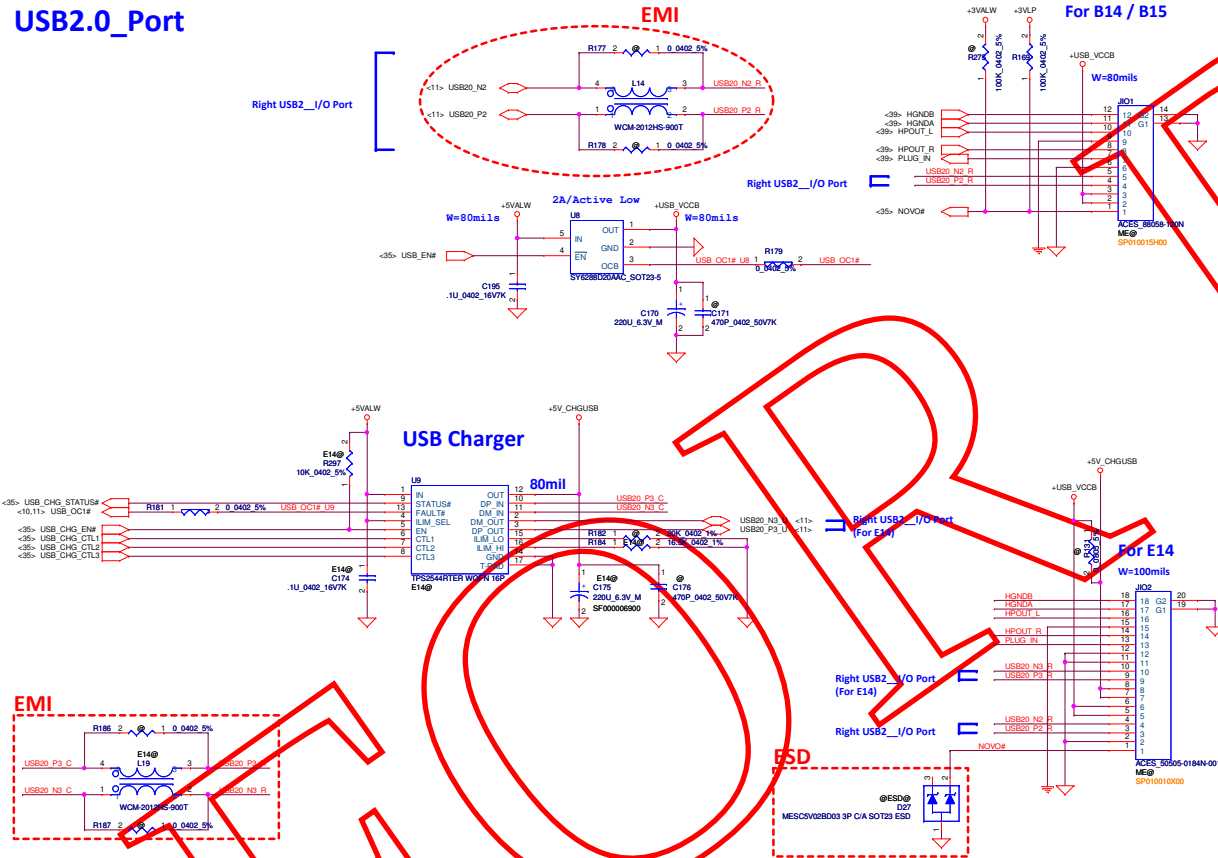
http://shop65127737.taobao.com



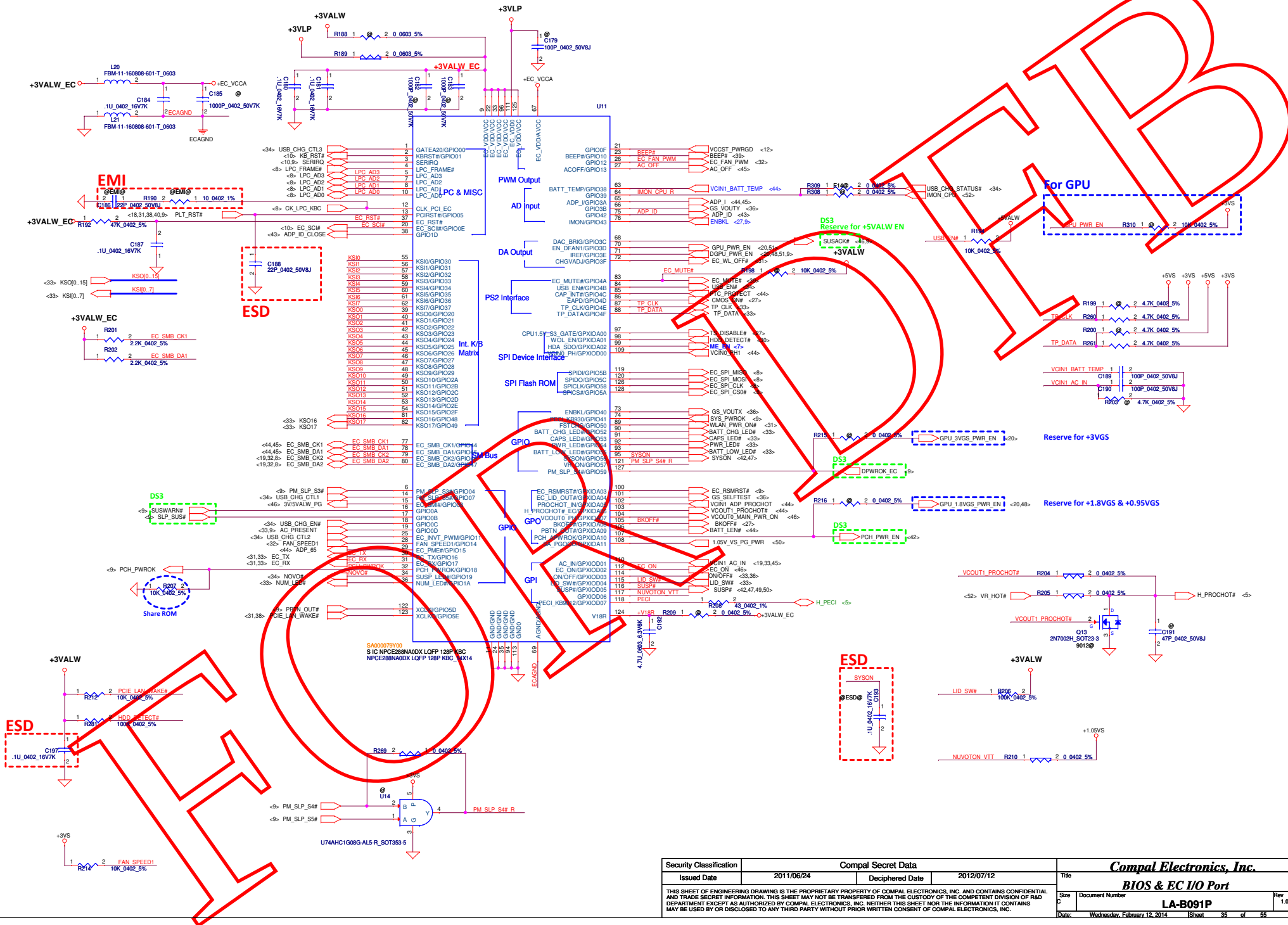
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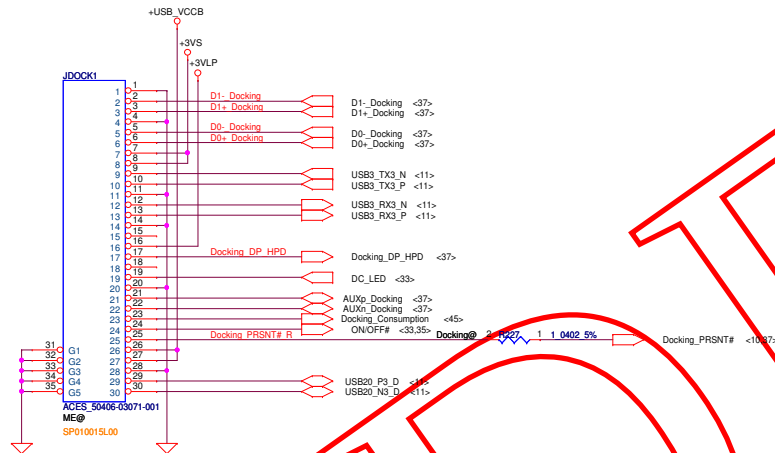
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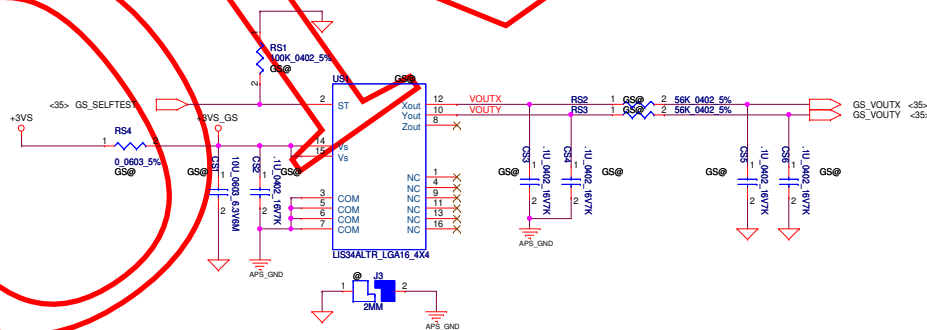
Security Classification	Compal Secret Data		Compal Electronics, Inc.	
Issued Date	2011/06/24	Deciphered Date	2012/07/12	File Name
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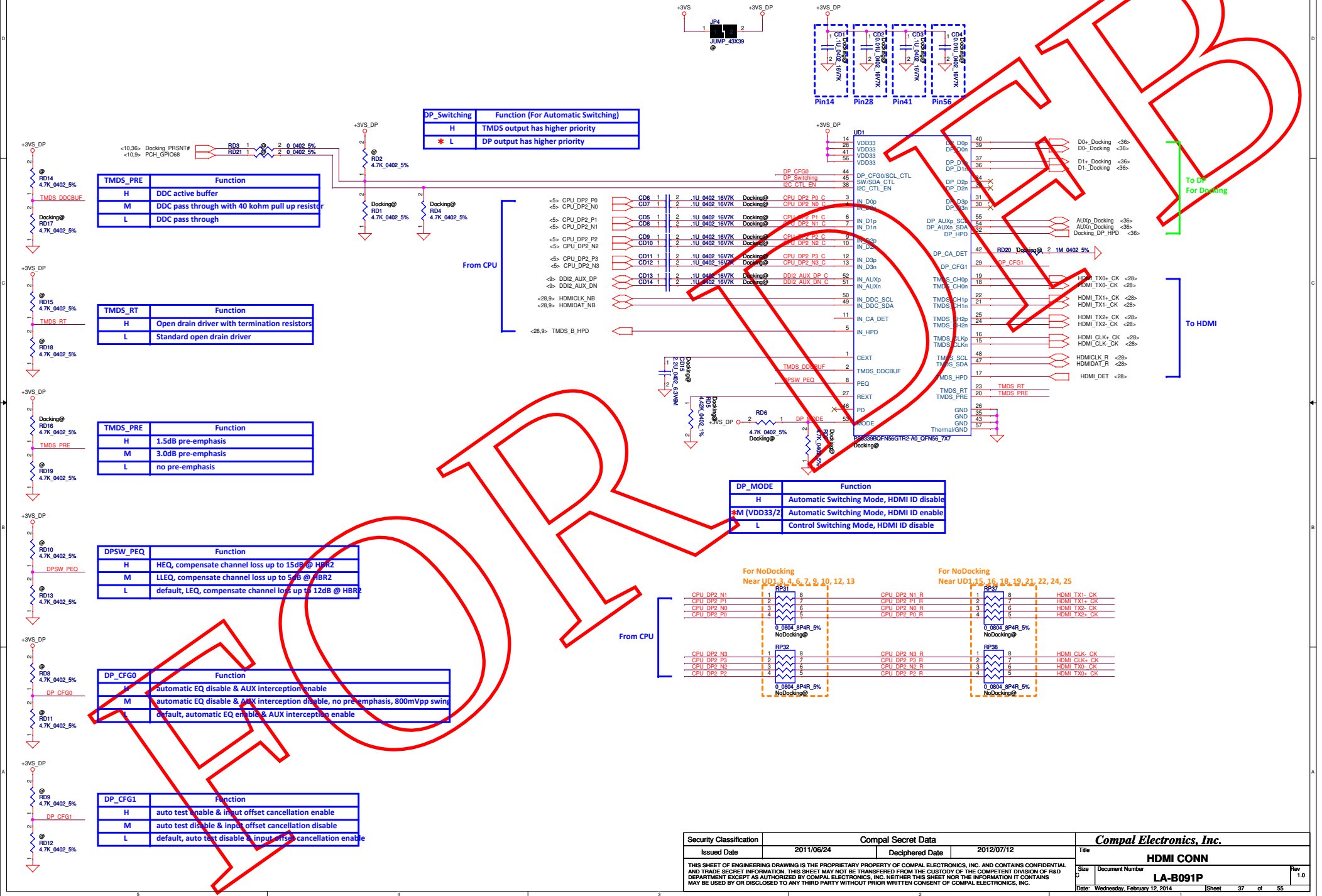


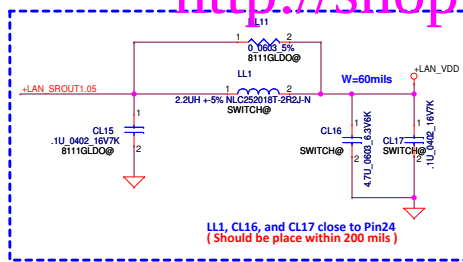
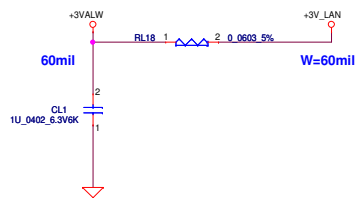
Security Classification	Compal Secret Data			Compal Electronics, Inc.	
Issued Date	2011/06/24	Deciphered Date	2012/07/12	Title	BIOS & EC I/O Port
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				Date	March二十六, 2014
				Version	98 of 66



APS (G-Sensor)





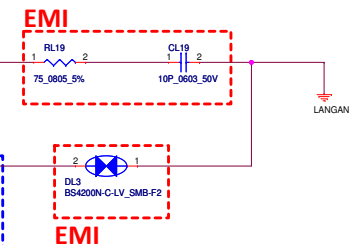
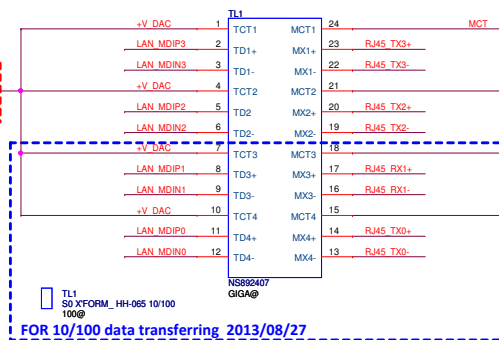
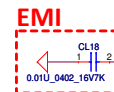
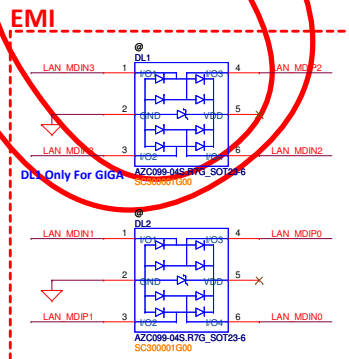
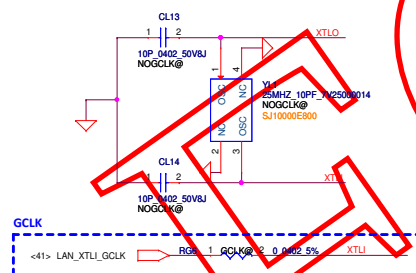
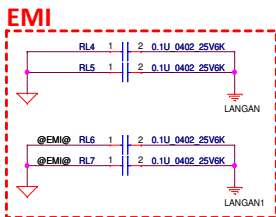
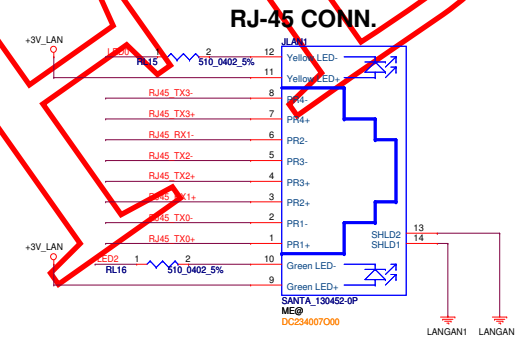
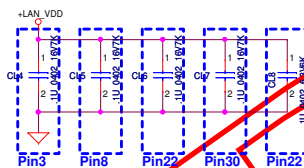
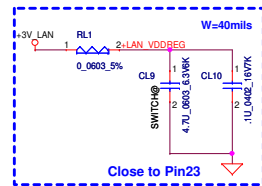
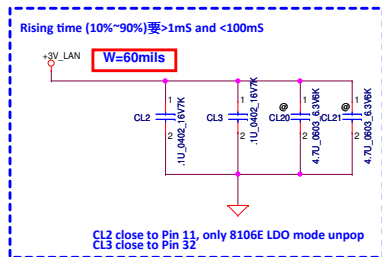


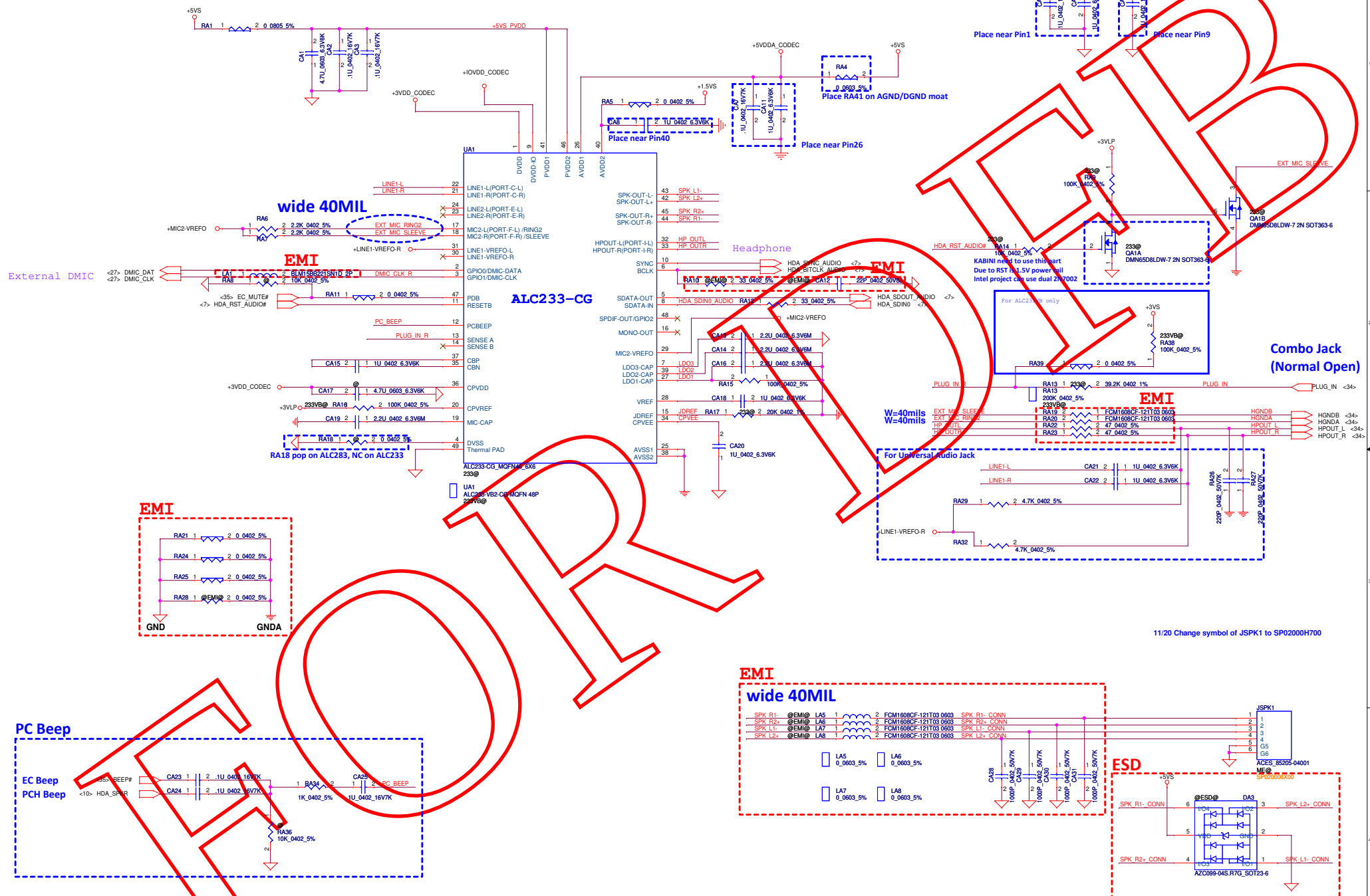
SA00005V700

SA000005Y00

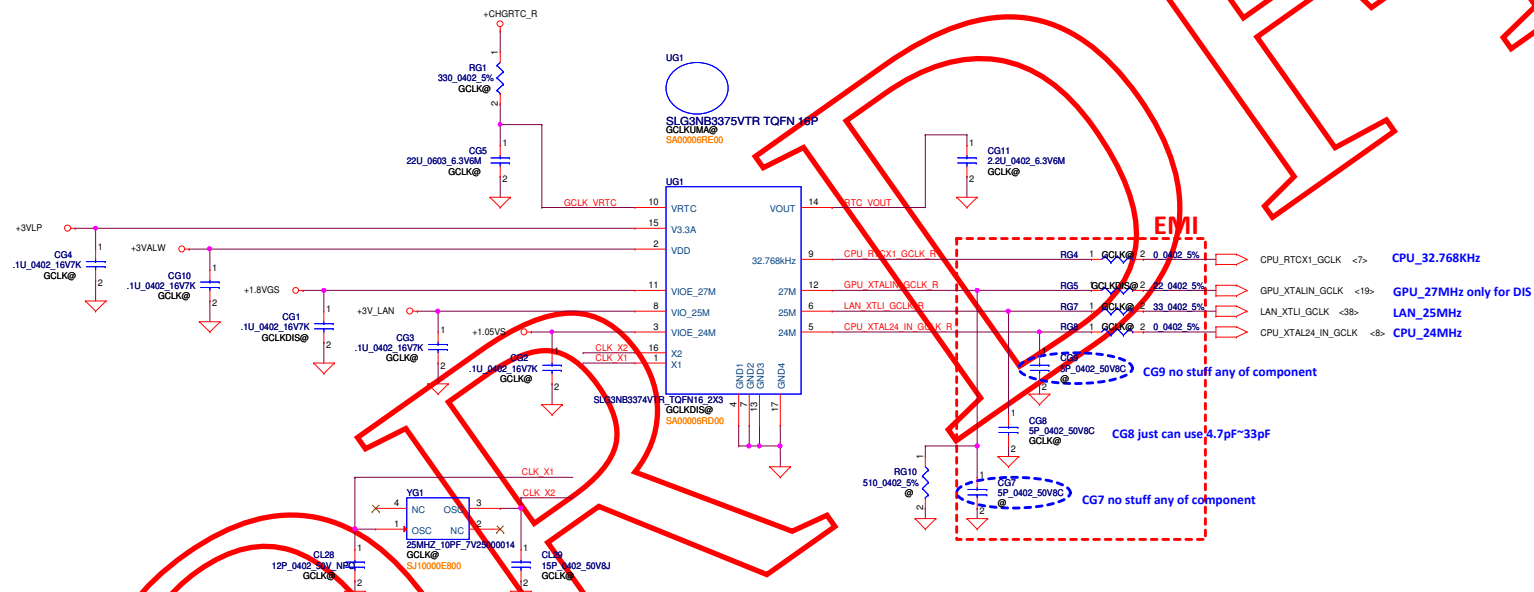
	1.0 V source	LL1	CL16, CL17	CL9, CL10	RL11	CL15
RTL8111G	LDO	X	X	X	O	O
RTL8111G	External	X	X	X	X	O
RTL8111GS/ RTL8111GUS/ RTL8106EUS	SWR	O	O	O	X	X
RTL8106E	LDO	X	X	X	X	X

Please refer to the table above when using different 1.0V supply source.
For RTL8111GS, RTL8111GUS, RTL8106E and RTL8106EUS, External 1.0V Supply Is Not Permitted.



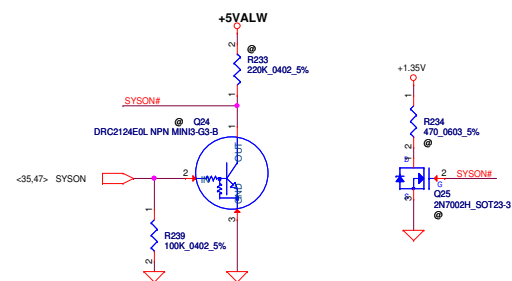
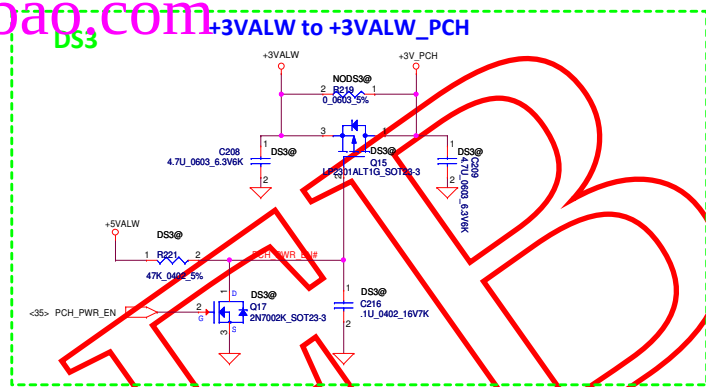


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Size	Document Number					Rev
C	LA-B091P					1.0
Date:	Wednesday, February 12, 2014			Sheet	39	of 55



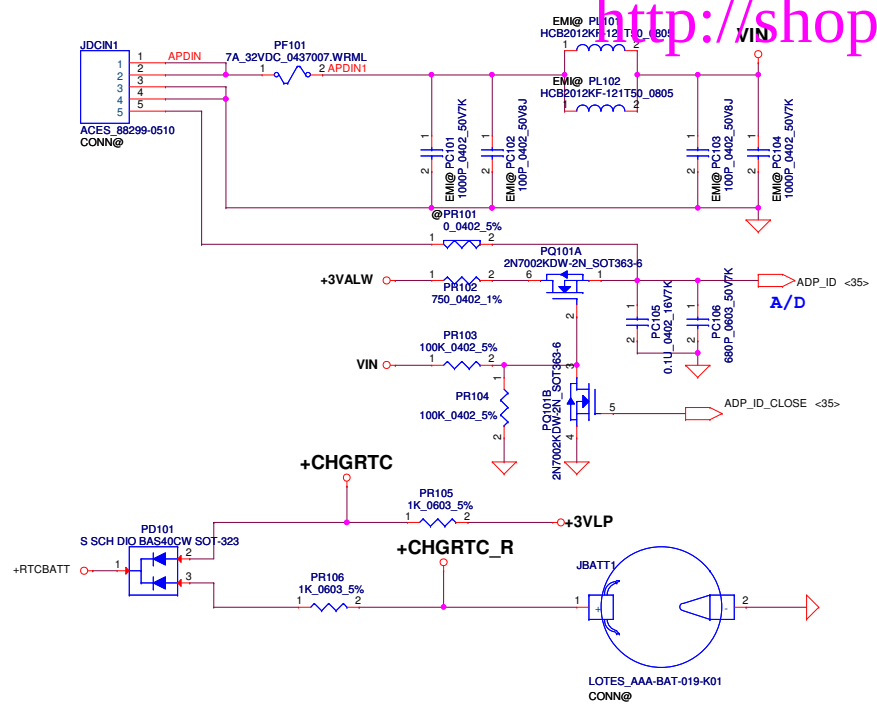
Security Classification		Compal Secret Data		Compal Electronics, Inc.	
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				Wednesday, February 12, 2014	Rev 1.0
				Sheet	41 of 55

DS3 +3VALW to +3VALW_PCH



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				Document Number LA-B091P	
Date:		Wednesday, February 12, 2014		Sheet	42 of 55

ADP_ID	AC Adapter	90W	65W
R(K ohm)	open	10	
ADP_ID(V)	3.3	1.65	
Detection voltage	>2.64	1.32~1.98	



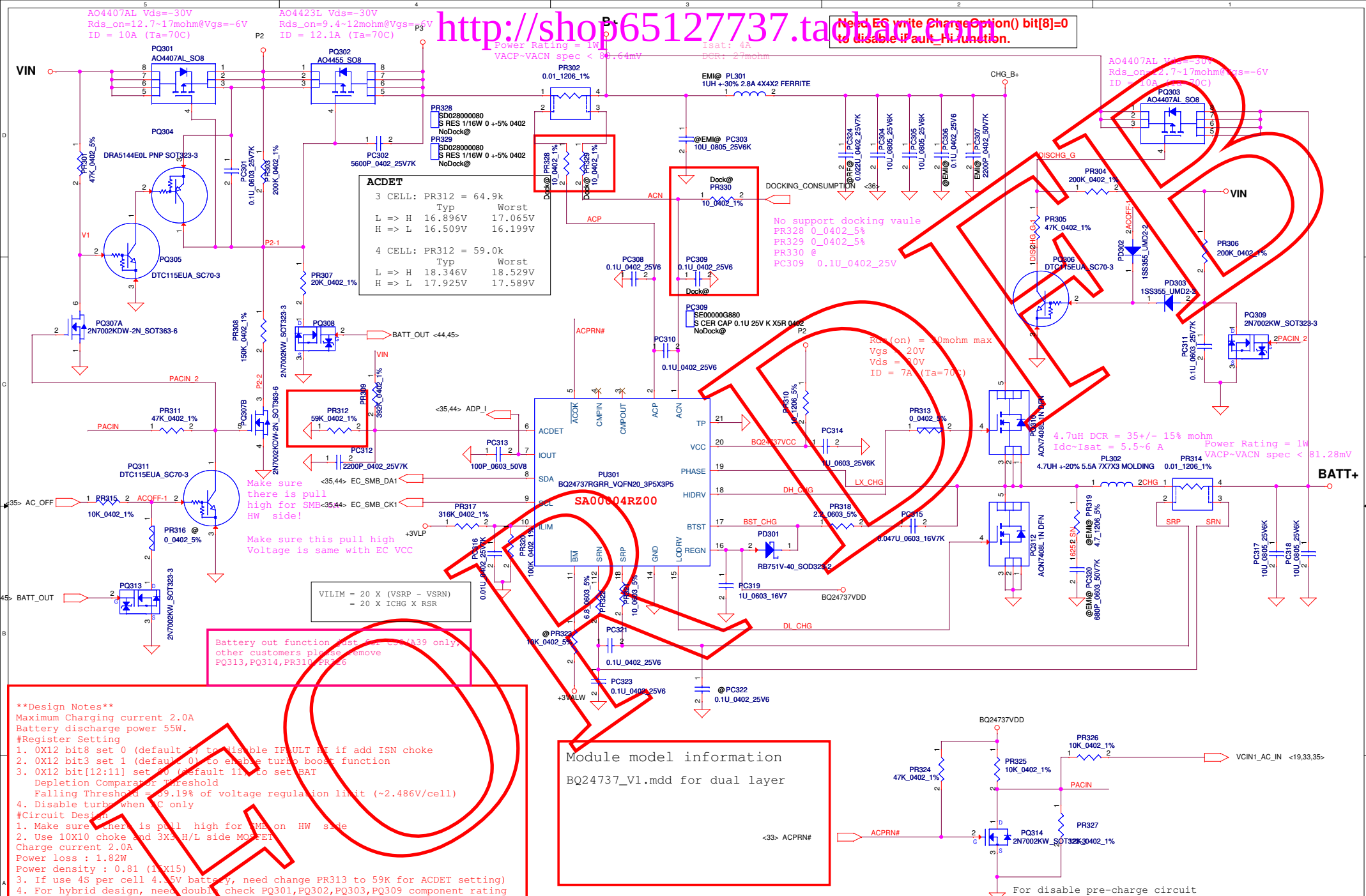
RTC Battery



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Need EC write ChargeOption() bit[8]=0 to disable ifault Hi function.



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					BE_BDW	1.0
				Date:	Wednesday, February 12, 2014	Sheet 45 of 55

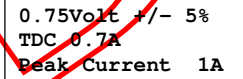
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EN1 and EN2 dont't floating

ENLDO_3V5V



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				Rev	1.0	
				Date:	Wednesday, February 12, 2014	Sheet 46 of 55



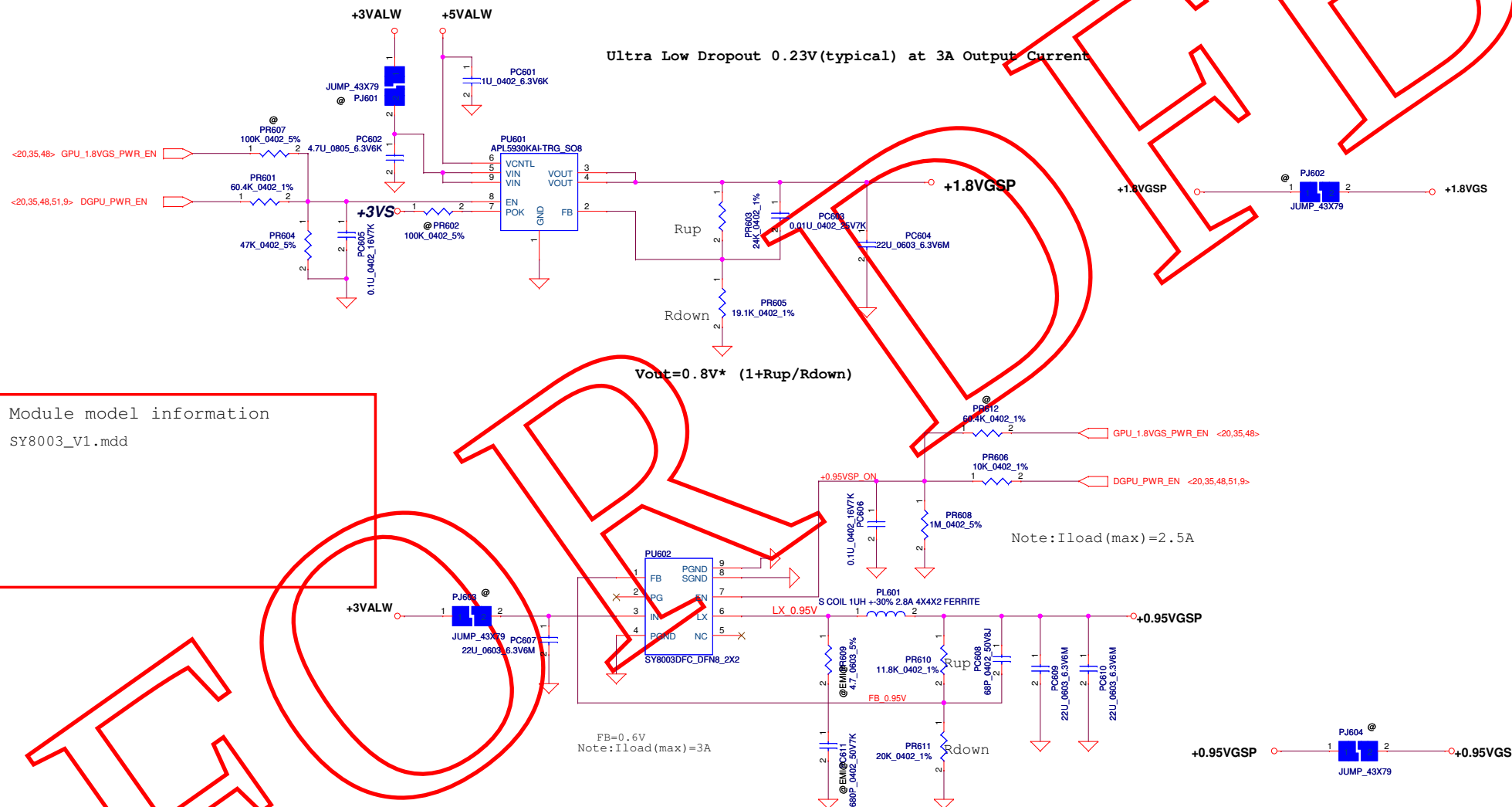
MOSFET footprint: SIS412DN

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				Custom	BE BDW	
Date: Wednesday, February 12, 2014				Sheet	47	of 55

Module model information

APL5930_V1.mdd

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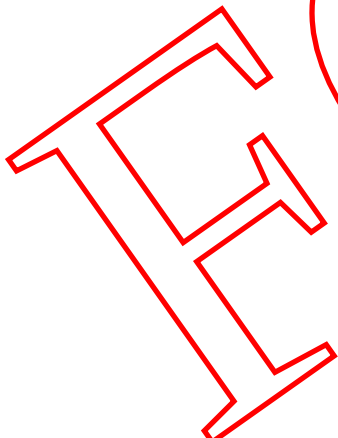


Module model information

SY8003_V1.mdd

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				Custom	1.0
				BE_BDW	
				Date:	Wednesday, February 12, 2014
				Sheet	48 of 55

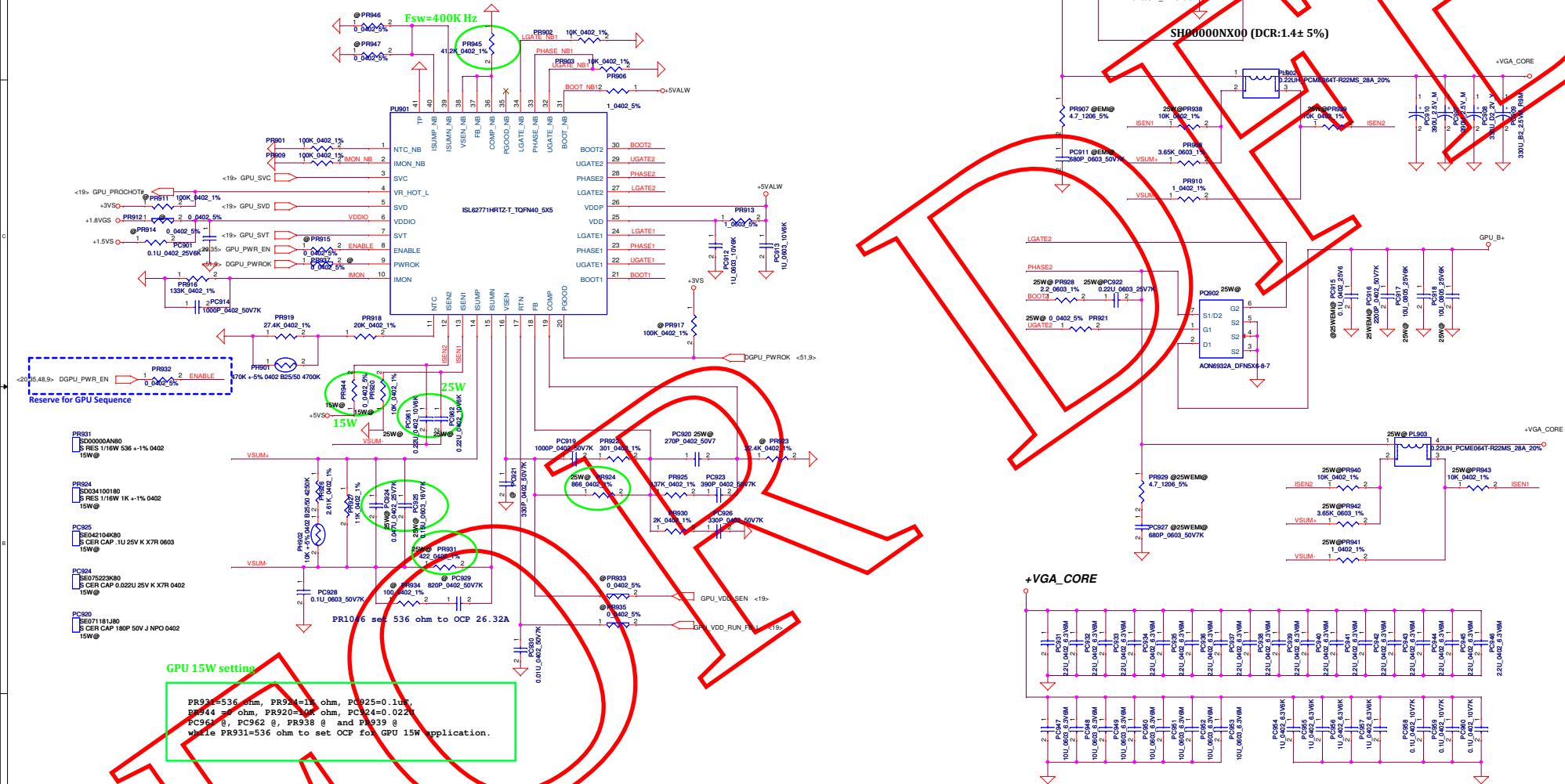
SY8208D_V1.mdd

[illegible]

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Module Model information
ISL62771_V1A.mdd for IC portion
ISL62771_V1B.mdd for SW portion

+VGA_CORE
AMD TOPAZ
TDC 31A, EDC 46.5A
OCP min 58.1A
AMD JET LE
TDC 20A, EDC 30A
OCP min 37.5A



GPU 15W setting

PR931=536 ohm, PR934=1k ohm, PC925=0.1uF,
PC944=0.1uF, PR920=300 ohm, PC924=0.022uF,
PC936=0.1uF, PC962=0.1uF, PR938=0.1uF, PC939=0.1uF,
while PR931=536 ohm to set OCP for GPU 15W application.

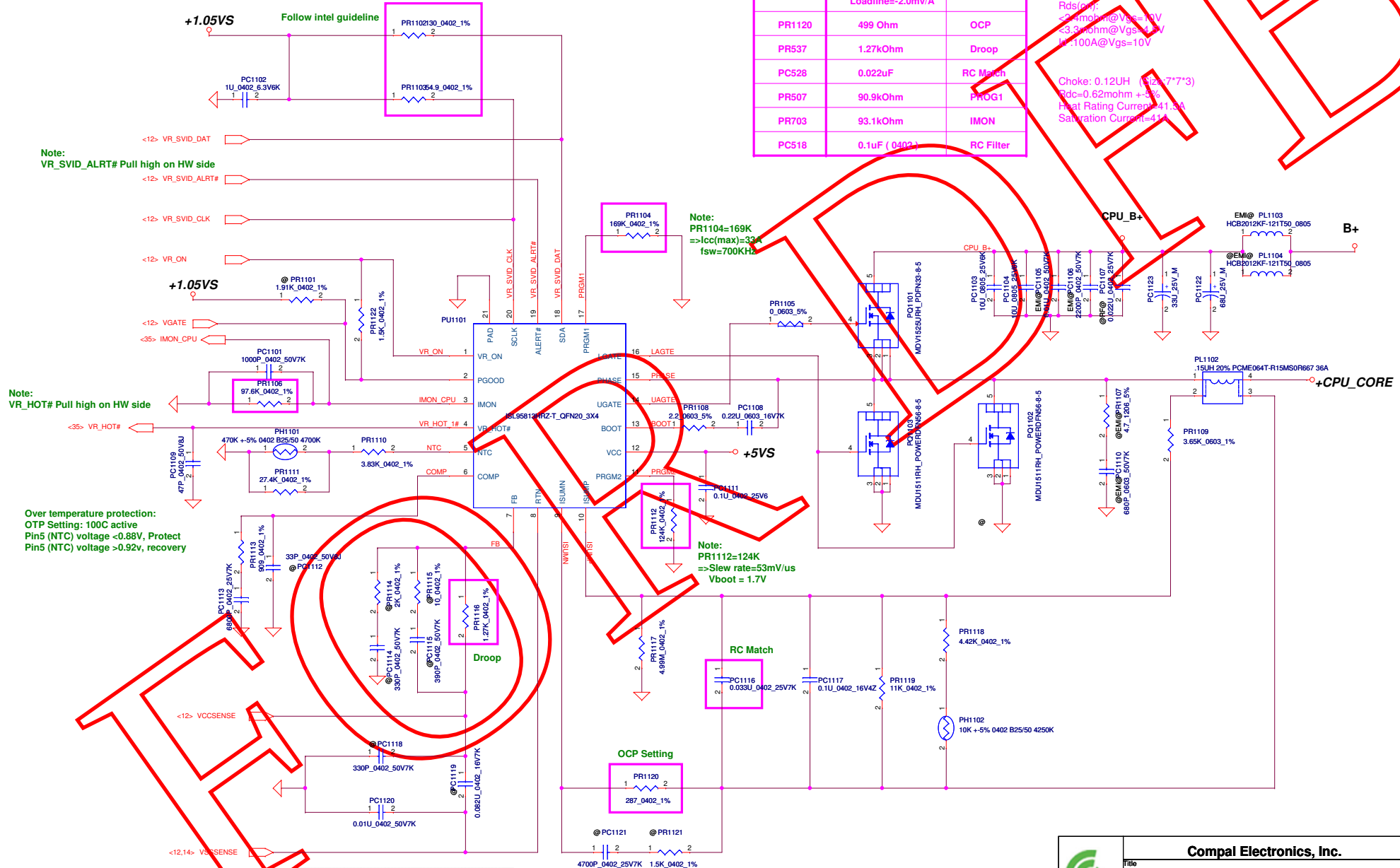
+VGA_CORE

Module model information:
ISL95813 (for 15W & 28W CPU)

Base on BDW PDDG Rev_0_73		
Location	15W	Note
	TDC 14A	
	MAX 32A	
	OCF 39A	
	Loadline=-2.0mv/A	
PR1120	499 Ohm	OCF
PR537	1.27kOhm	Droop
PC528	0.022uF	RC Match
PR507	90.9kOhm	PROG1
PR703	93.1kOhm	IMON
PC518	0.1uF (0402)	RC Filter

L-side MOS: MDU1511RH
Rds(on):
<2.4mohm@Vgs=10V
<3.3mohm@Vgs=4.5V
Id :100A@Vgs=10V

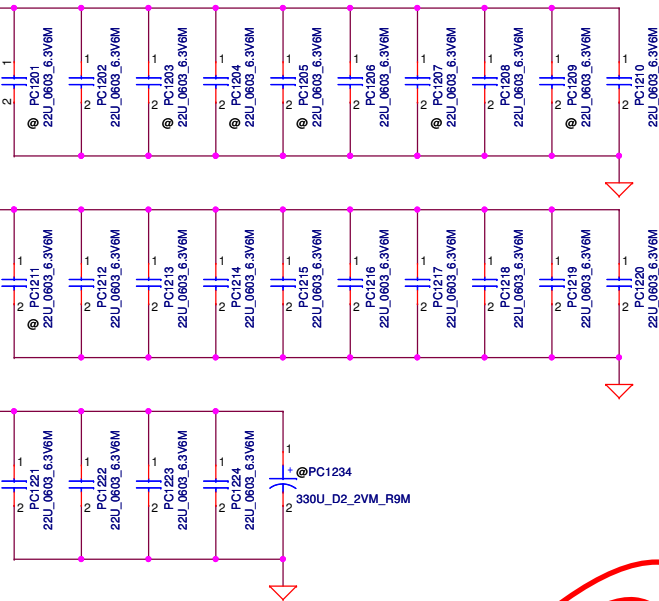
Choke: 0.12UH (Size:7*7*4)
Rdc=0.62mohm +-5%
Heat Rating Current=41.3A
Saturation Current=41A



Local sense put on HW site

+CPU_CORE

24 X 22u/0603



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Date: Wednesday, February 12, 2014		Sheet 53 of 55			

Item	Reason for change	PG#	Modify List	Date	Phase
1					
2					
3					
4					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					

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Date: Wednesday, February 12, 2014				Document Number BE BDW
Sheet 54 of 55				Rev 1.0

Item	Page	MODIFICATION LIST	PURPOSE	EVT TO DVT
1	P. 36	Modify DP_SEL schematic	Because the first design is wrong.	
2	P. 34	Delete D28	It already reserve in sub BD	
3	P. 36	Modify HPD schematic	Because the first design is wrong.	
4	P. 36	Modify DP AUX schematic	Cap already reserve in sub BD	
5	P. 20	Reserve +1.05VS to +0.95VGS	AMD's suggestion	
6	P. 33	Add D26 for ESD		
7	P. 42	Add RV198, RV199	AMD's suggestion	
8	P. 22-24	Add GPU Termination Resistance	AMD's suggestion	
----- DVT TO PVT -----				
1	P. 35	change U11.111 power rail to +3VLP	It only use +3VLP	
2	P. 33	un-pop R294, pop R295.	B series's LED need to follow E series	
3	P. 10	Add R247, R248	For BIOS Stap Pin	
4	P. 20	Add RV60, delete RV36	for GPU Sequence	
5	P. 20	Add RV61, delete RV240	for GPU Sequence	
6	P. 37	Change DP Switch IC solution	For HDMI audio issue	
7	P. 35	Add C197 for ESD		
8	P. 33	Add C198 for ESD		
9	P. 30	Add C199 for ESD		
----- PVT TO PRE-MP -----				
1	P. 33	Reserve R298, R299 for DC-in LED control	To avoid LED shimmer	
2	P. 38	Change DL1 and DL2 footprint for ESD		